

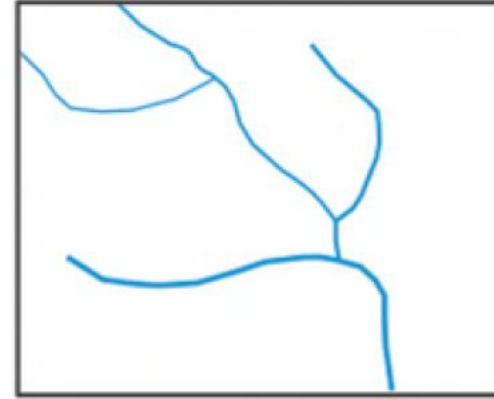
QUANTITATIVES MEASURES OF SPACE

Components of spatial entities

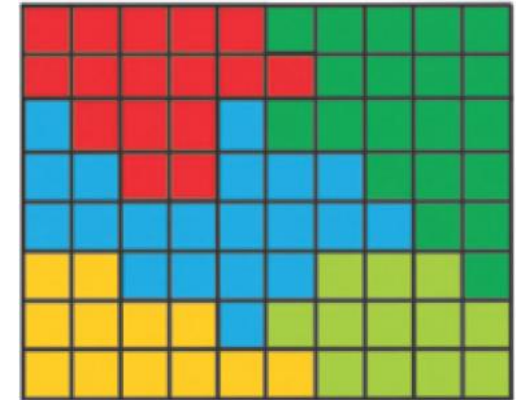
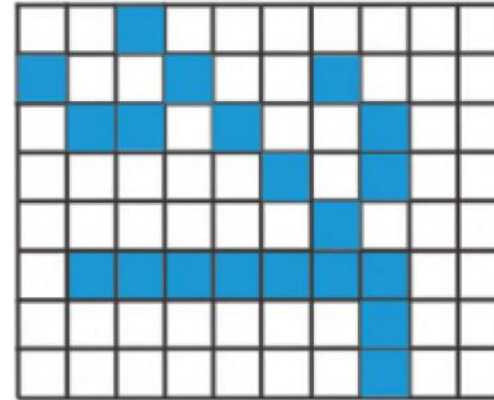
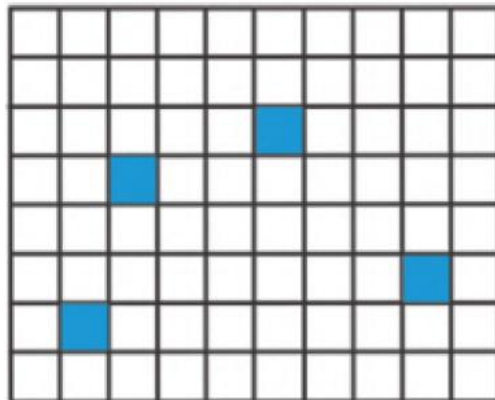
- Easting (E)
- Northing (N)
- Height or elevation above the reference geodatum (ht)
- Attribute at the given location (M)
- Time of observation (t)

Object and field view of spatial entities

Object view
(vector feature)



Field view
(raster feature)



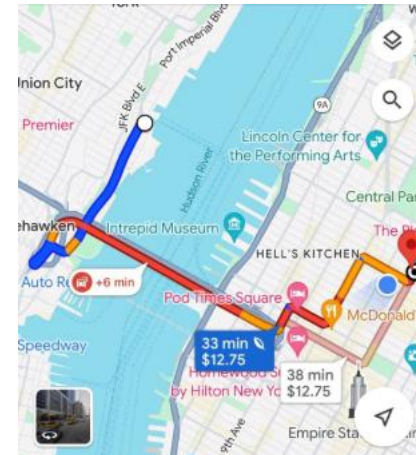
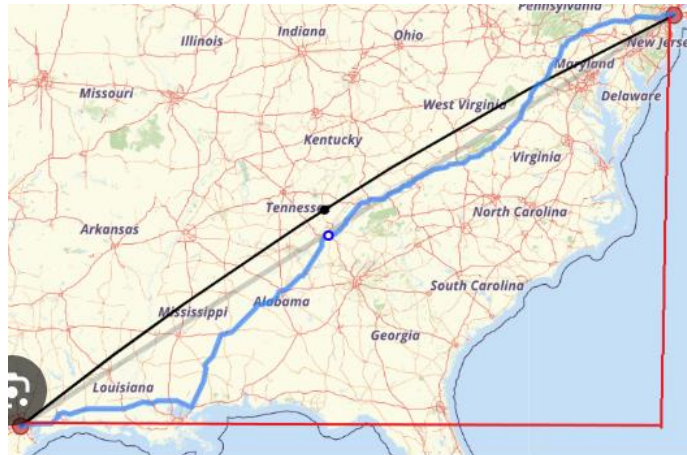
Object and field view of spatial entities- A comparison

Item	Object view	Field view
Philosophy of representation	Continuous 2D space, location explicit; representation of multiple attributes in one layer possible, with more columns in the tabular format	Discrete 2D space comprising regular grids of same size and shape; location implicit; one attribute per layer
Accuracy of representation	Precise, objects visible	Crude, accuracy subject to grid cell size; no objects, only pixels of different attribute values
Data volume	Compact, but overhead essential	Bulky, inefficient, and subject to grid cell size
Ease of updating	Complex (spatial component and attributes must be updated simultaneously)	Easy (only the concerned layer needs attention)
Best uses	Spatially discontinuous entities (e.g., roads); descriptive analysis; certain kinds of modelling requiring precise geometric properties (e.g., fire fronts)	Spatially varying attributes continuously; spatial modelling

Quantitative measures of space

- Distance
- Proximity
- Neighborhood
- Dispersion and clustering
- Direction
- Area

Distance



Distance types

- Euclidean distance
- Geodesic distance
- Manhattan distance
- Network (or Cost) distance
- Surface distance
- Minkowski distance

Euclidean distance

- Shortest distance between two points in a Cartesian plane

$$D_{Euclidean} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

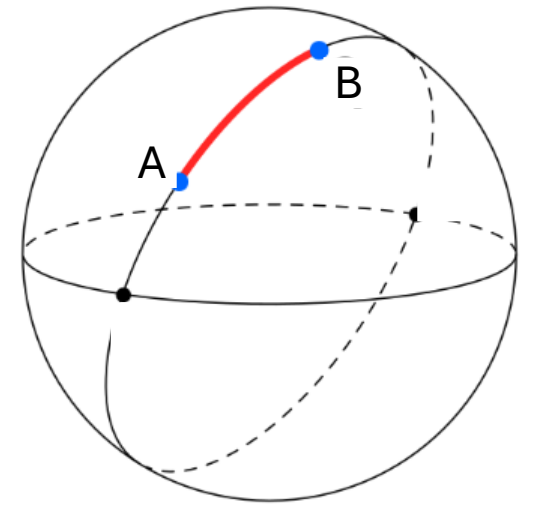
Geodesic distance

- Distance across the curved surface of the world
- Calculated using the haversine formula
- Difference between euclidean and geodesic distance increases proportionally with distance from the source

$$D_{Geodesic} = R \cdot c$$

$$c = 2 \times \operatorname{atan2}(\sqrt{a}, \sqrt{1-a})$$

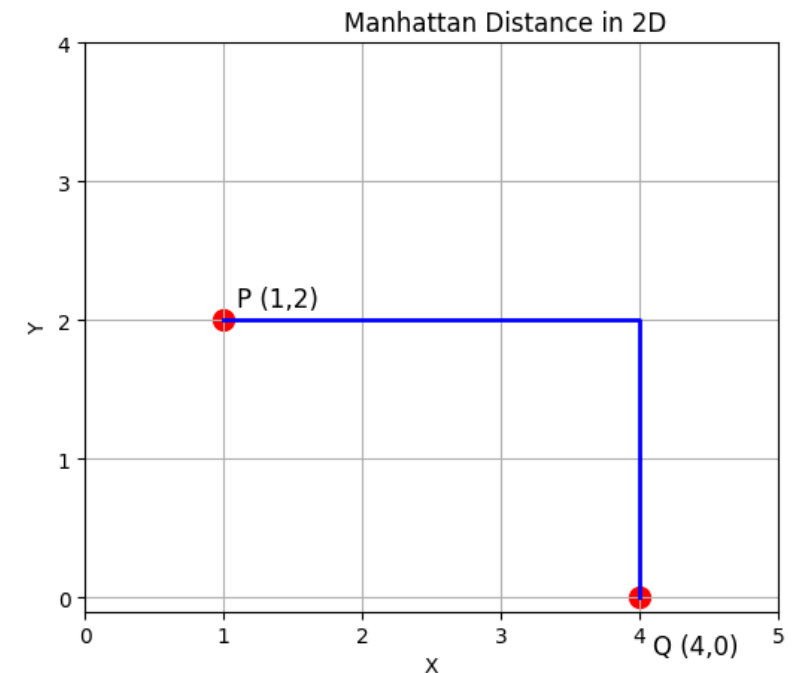
$$a = \sin^2\left(\frac{\phi_B - \phi_A}{2}\right) + \cos\phi_A \cos\phi_B \sin^2\left(\frac{\lambda_B - \lambda_A}{2}\right)$$



Manhattan distance

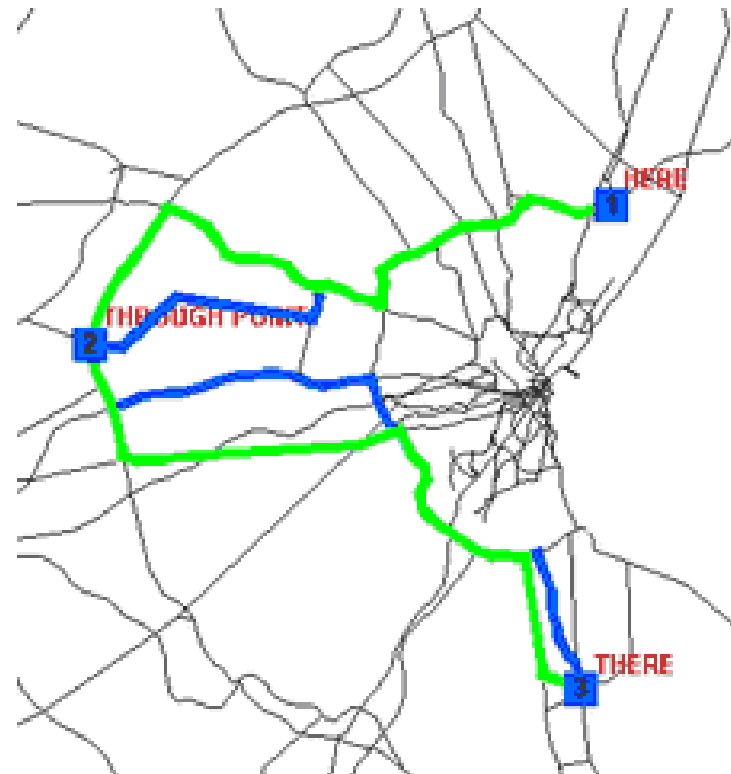
- Measures how far apart two points are by summing the absolute differences of their coordinates
- Unlike Euclidean distance, it calculates distance along grid-like paths
- Also known as L1 or taxicab distance

$$D_{Manhattan} = |x_2 - x_1| + |y_2 - y_1|$$



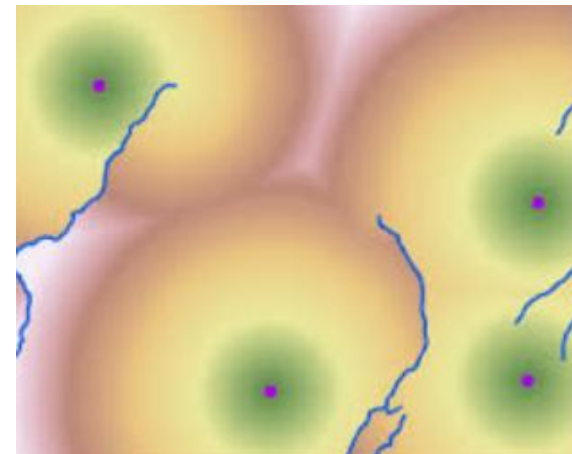
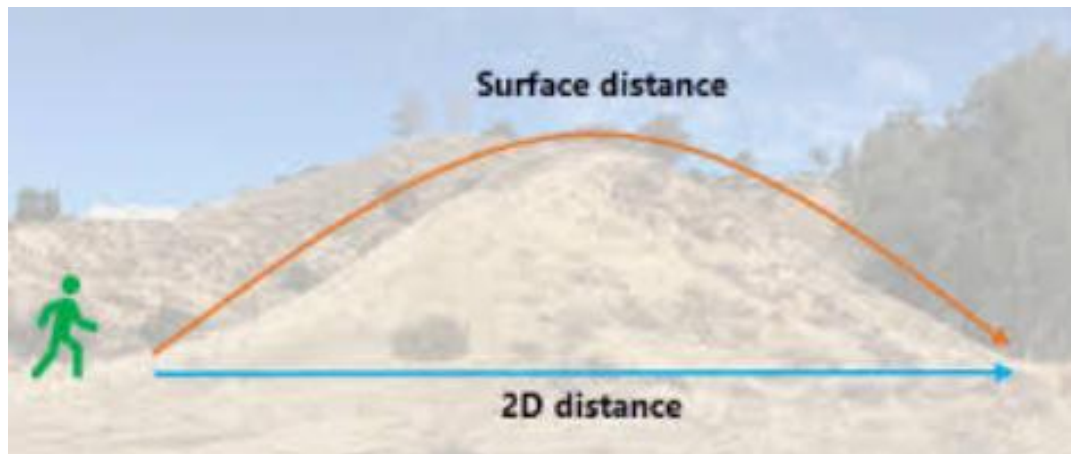
Network (or Cost) distance

- Distance (or cost, time etc.) involved in traversing a network
- Determine the shortest weighted distance (or accumulated travel cost) from source to a destination



Surface distance

- Average distance between the boundaries of a segmented region and the ground truth
- Provides a measure of overall boundary localization accuracy
- Used in precise anatomical boundary localization tasks



Minkowski distance

- A generalization of both the Euclidean distance and the Manhattan distance

$$D_{Minkowski} = (|x_2 - x_1|^p + |y_2 - y_1|^p)^{\frac{1}{p}}$$

Proximity

- Spacing among multiple spatial entities or objects within a given space domain
- Data samples in close proximity tend to have shorter distances between them
- Lack of proximity is manifested by a large distance between neighboring observations and indicates a dispersed distribution

1



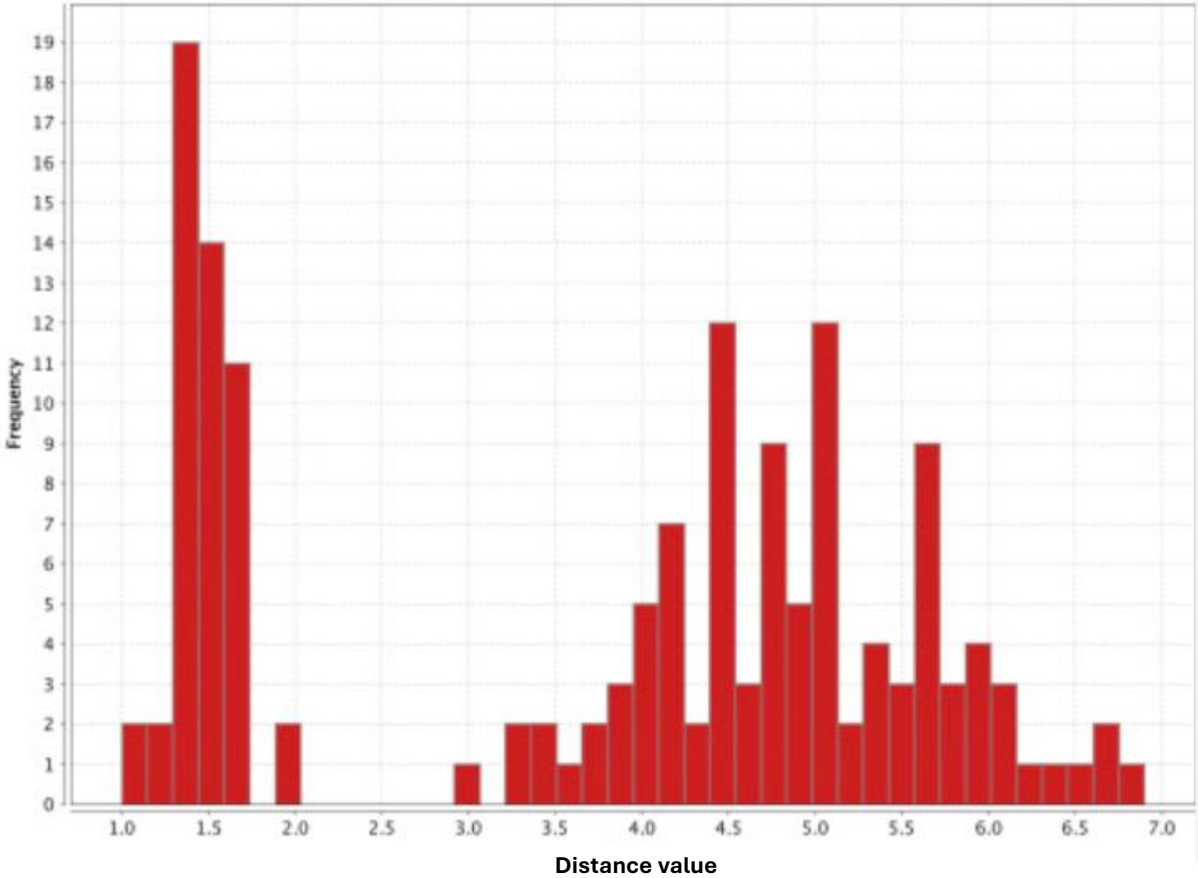
1

[illegible][illegible]

Distance matrix

	1	2	3	4	5	6	7	8	9	10
1	$d_{1,1}$	$d_{1,2}$	$d_{1,3}$	.	.	.	.	.	.	$d_{1,10}$
2	$d_{2,1}$	.	.	.	.	.	.	.	.	$d_{2,10}$
3	$d_{3,1}$	.	.	.	.	.	.	.	.	$d_{3,10}$
4	.	.	.	.	.	.	.	.	.	.
5	.	.	.	.	.	.	.	.	.	.
6	.	.	.	.	.	.	.	.	.	.
7	.	.	.	.	.	.	.	.	.	.
8	.	.	.	.	.	.	.	.	.	.
9	.	.	.	.	.	.	.	.	.	.
10	$d_{10,1}$	$d_{10,2}$	$d_{10,3}$	.	.	.	.	.	.	$d_{10,10}$

Histograms of distances for
a dispersed distribution and
a close proximity distribution



Proximity toolset in ArcGIS

“What is near to what?” – one of the basic questions asked of a GIS

- How close is a well to a landfill?
- Do any roads pass within 1,000 meters of a stream?
- What is the distance between two locations?
- What is the nearest or farthest feature from something?
- What is the distance between each feature in a layer and the features in another layer?
- What is the shortest street network route from some location to another?

Proximity toolset in ArcGIS

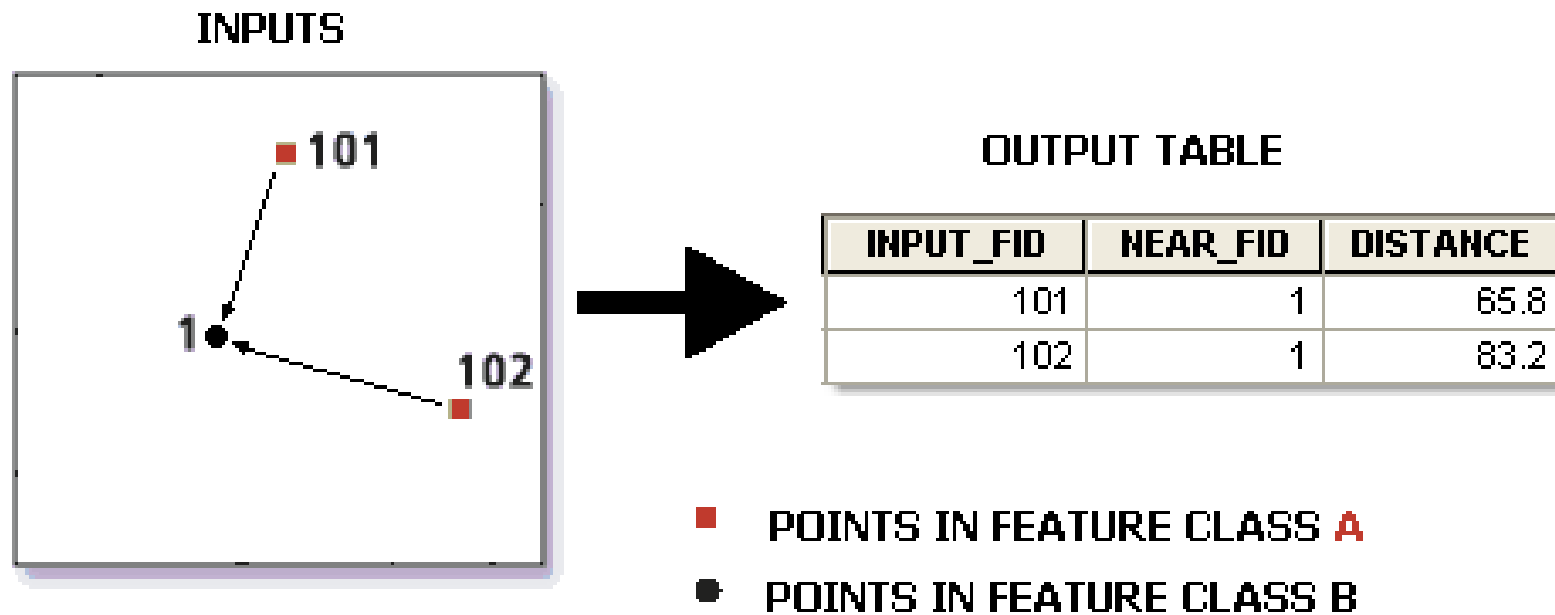
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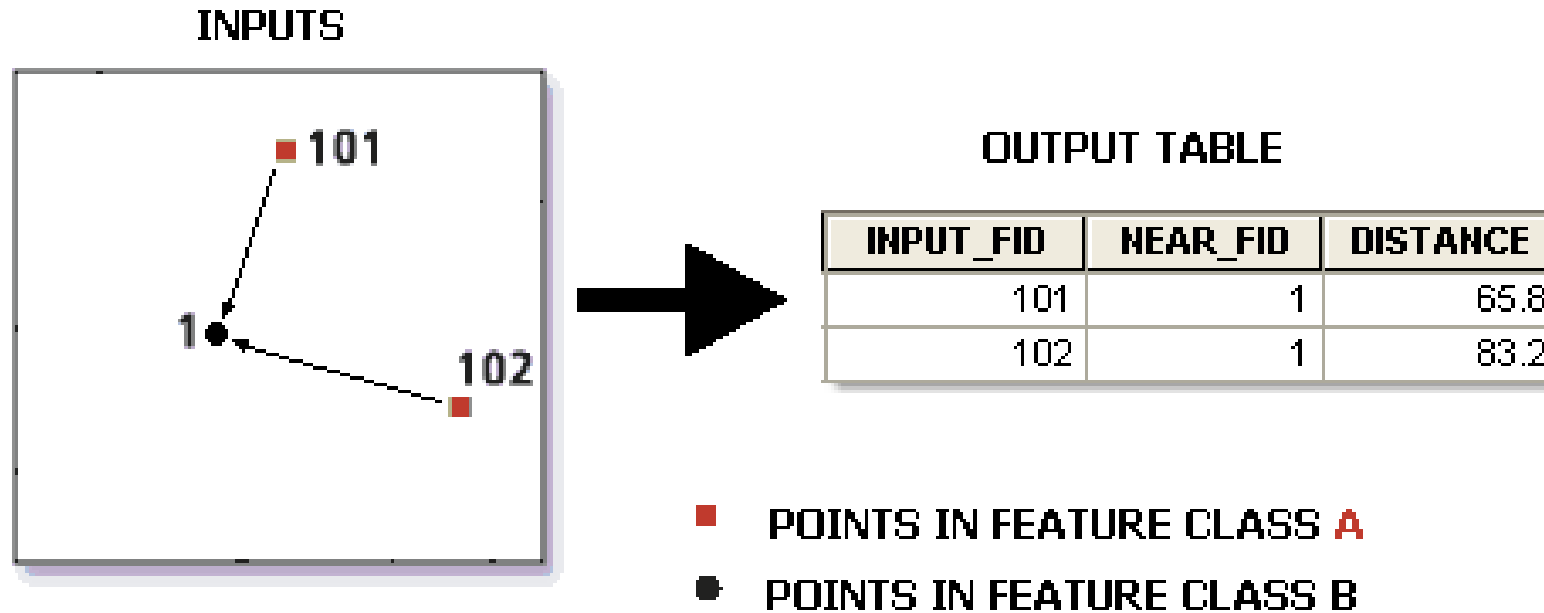
- How close is a well to a landfill?
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- Proximity toolset contains tools that are used to determine the proximity of features within one or more feature classes or between two feature classes
- Proximity toolset contains tools that:
 - Identify features that are closest to one another
 - Calculate the distances between or around them

Point distance tool (of Proximity toolset)

Determines the distances from input point features to all points in the near features within a specified search radius





- The tool creates a table with distances between two sets of points. if the default search radius is used, distances from all input points to all near points are calculated. The output table can be quite large. For example, if both input and near features have 1,000 points each, then the output table can contain one million records.
- Use a meaningful search radius to limit the size of the output and improve tool performance. The output table contains only those records that have a near point within the search radius.

Neighborhood

- Spatial extent or range within which a spatial entity exists or should be considered
- Defined in terms of continuity, distance and weights

Neighborhood perceived from object view and field view of representation

Object view

- Distance-based, defined by a radius, resulting in the neighborhood always having a circular shape. Uniform distance is applied to all qualifying neighbors.
- Shape can be made oval by defining the neighborhood by two radii, one along the planar axis and another in the mutually perpendicular planar axis.
- A circular neighborhood means a uniform distance is applied to all qualifying neighbors, while an oval neighborhood is defined by two distances.
- Neighborhood centres at the point in question, and the radius length represents the neighborhood size.
- All observations falling within the defined spatial extent (e.g., inside the circle or oval) are considered as neighbors.

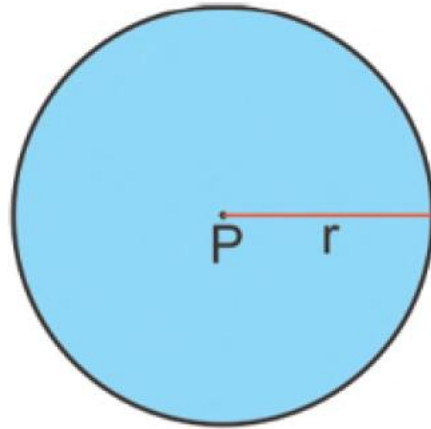
Field view

- Neighborhood involves both distance and adjacency.
- In terms of distance, neighborhood can be defined by the window size surrounding the cell.
- Based on adjacency, neighborhood falls into two types: 4- and 8-connectivity in the raster format.

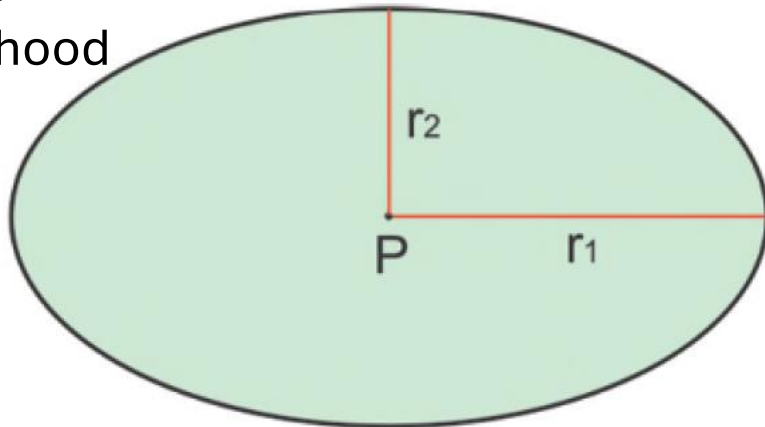
Neighborhood perceived from object view and field view of representation

Object view of representation

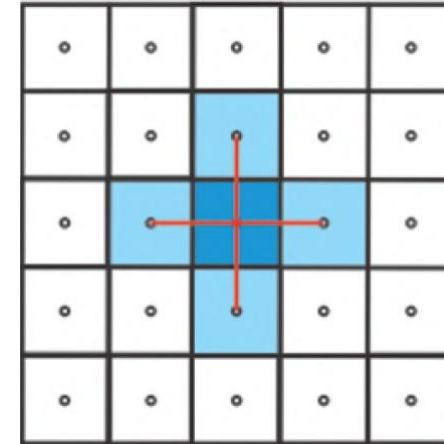
Circular neighborhood



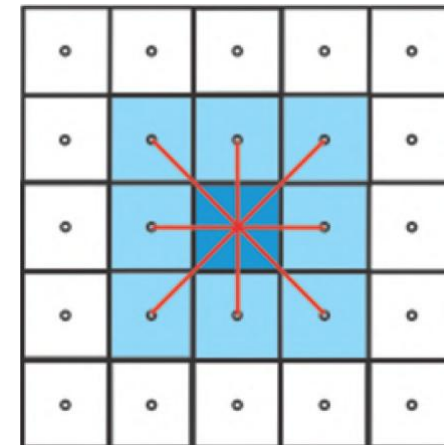
Oval neighborhood



Field view of representation



4-connectivity neighborhood

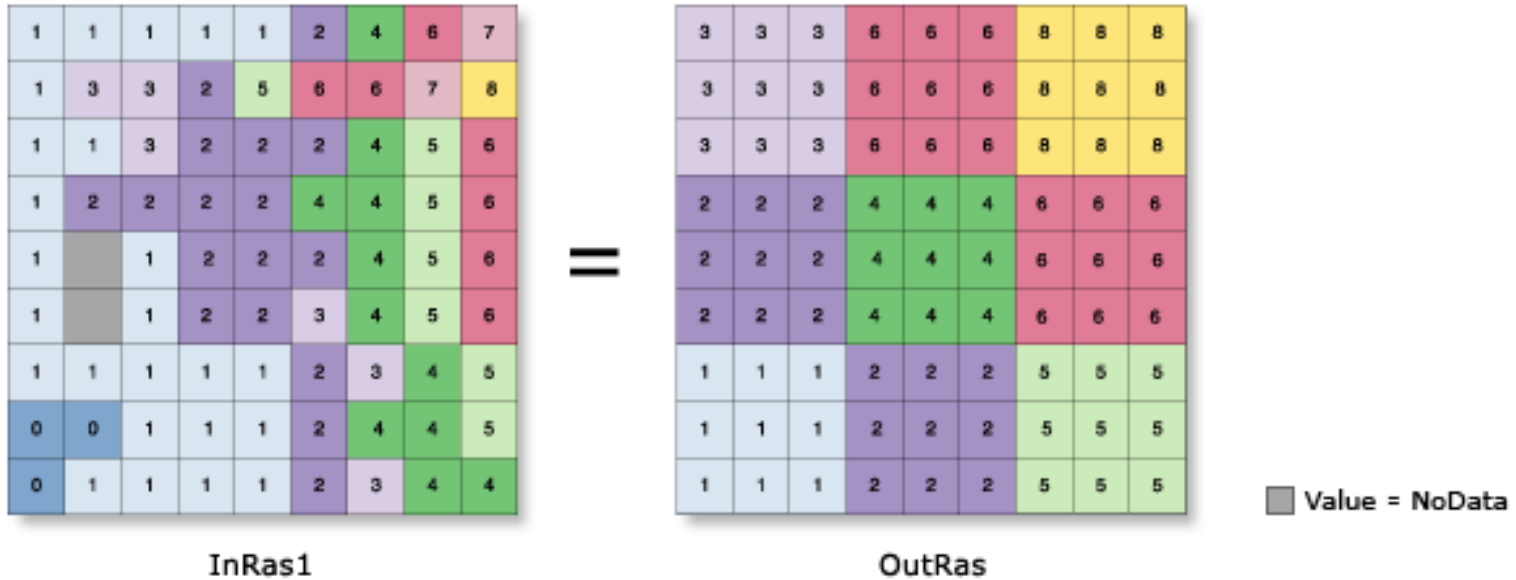


8-connectivity neighborhood

Neighborhood toolset in ArcGIS

- Creates output values for each cell location based on the location value and the values identified in a specified neighborhood
- Neighborhood type can be moving radius or search radius
- Tools available in the neighborhood toolset:
 - Block statistics
 - Filter
 - Focal Flow
 - Focal Statistics
 - Line Statistics
 - Point Statistics

Block statistics



OutRas = BlockStatistics(InRas1, NbrRectangle(3,3,"CELL"), "MAXIMUM", "DATA")

Focal statistics

0	1	1	0	3	0
0	3	3	2	0	
4	0	3	2	4	2
0	2	2	1	4	2
0		0	4	4	0
0	2	0	0	0	3

InRas1

=

4	8	10	9	5	3
8	15	15	18	13	9
9	17	18	21	17	12
6	11	14	24	23	16
4	6	11	15	18	13
2	2	6	8	11	7

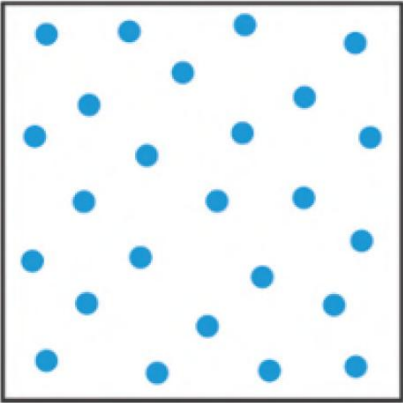
OutRas

 Value = NoData

OutRas = FocalStatistics(InRas1, NbrRectangle(3,3,"CELL"), "SUM", "DATA")

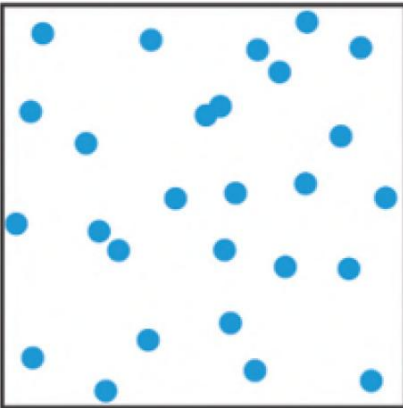
Dispersion and clustering

- Dispersion refers to the wide scattering of observations over the area of study
- Clustering refers to the spatial grouping of certain observations that are much closer to each other than to the members of another group



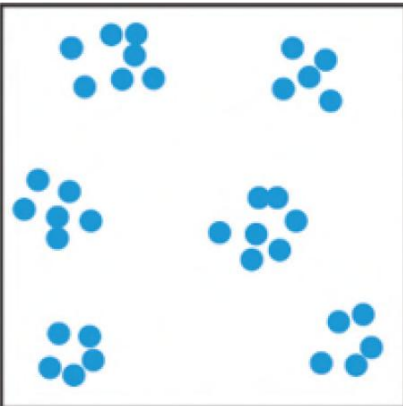
Uniform dispersion

If all observations are widely scattered with a roughly equal distance between the nearest neighbors, they are said to be uniformly dispersed in their distribution



Random dispersion

If the distance between the nearest neighbors is variable, then the pattern can be characterized as random dispersion



Clumped dispersion

If there is a huge variation in the distance between the nearest neighbors, some of them tend to be in close proximity to one another, much further away than to the members of another spatial group is variable, then the pattern is characterized as clumped dispersion

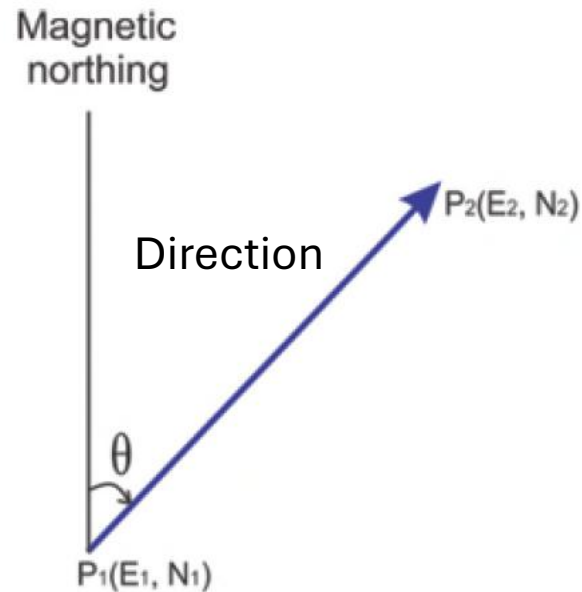
Direction

- Orientation or bearing in which an object moves or heads
- Commonly associated with linear features or topographic surfaces
 - Linear features: Indicative of the bearing to which an object is moving. E.g., water in a river channel flowing from upstream to downstream
 - Surface features: Air pollutants dispersed from the source to the nearby area of a lower concentration along the predominant wind direction

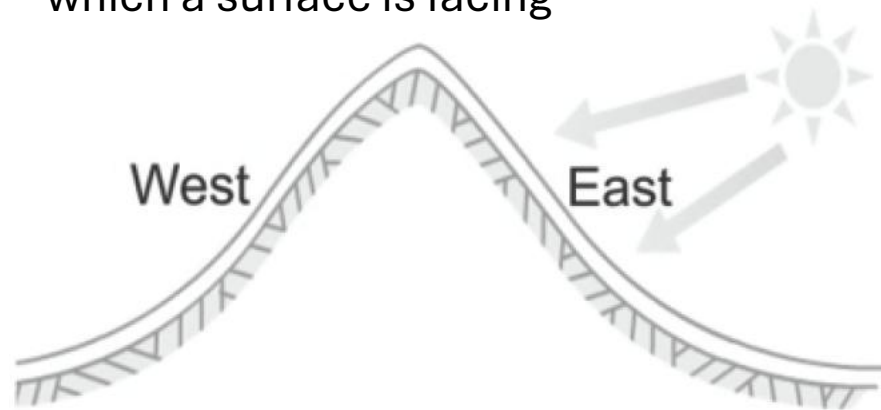
Direction (contd.)

For object view of representation

- Direction is referenced from true magnitude north clockwise
- Direction of motion is determined by the relativity of the starting and ending positions in the local coordinate system



Orientation: Direction to which a surface is facing

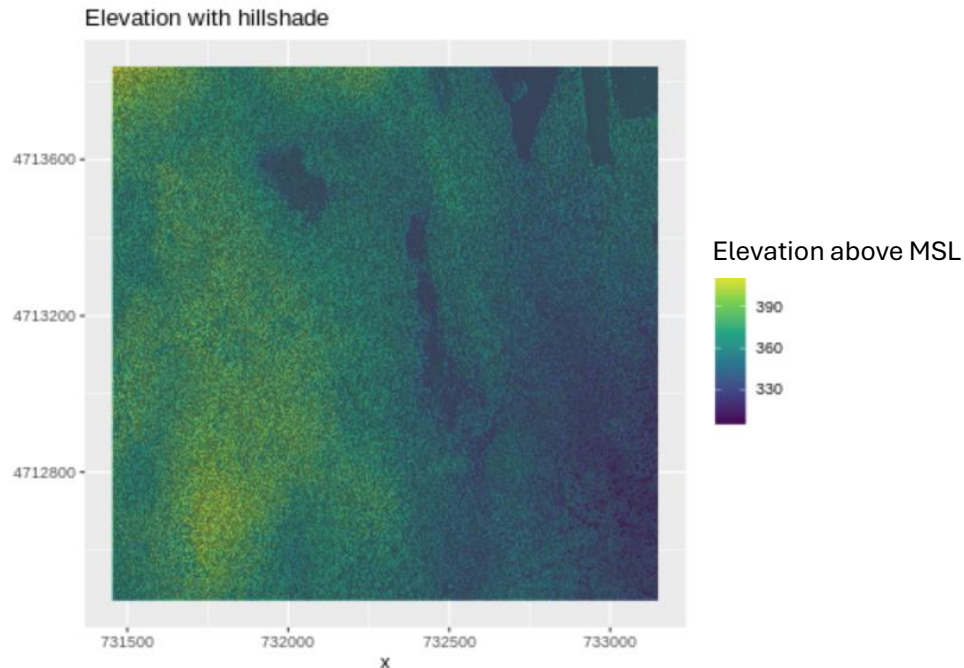


$$\theta = \tan^{-1} \left(\frac{\Delta N}{\Delta E} \right)$$
$$= \tan^{-1} \left(\frac{N_2 - N_1}{E_2 - E_1} \right)$$

Direction (contd.)

For field view of representation

- Direction is determined from the observed height of grid cells for a defined neighborhood
- Deals with the aspect to which a slope is facing

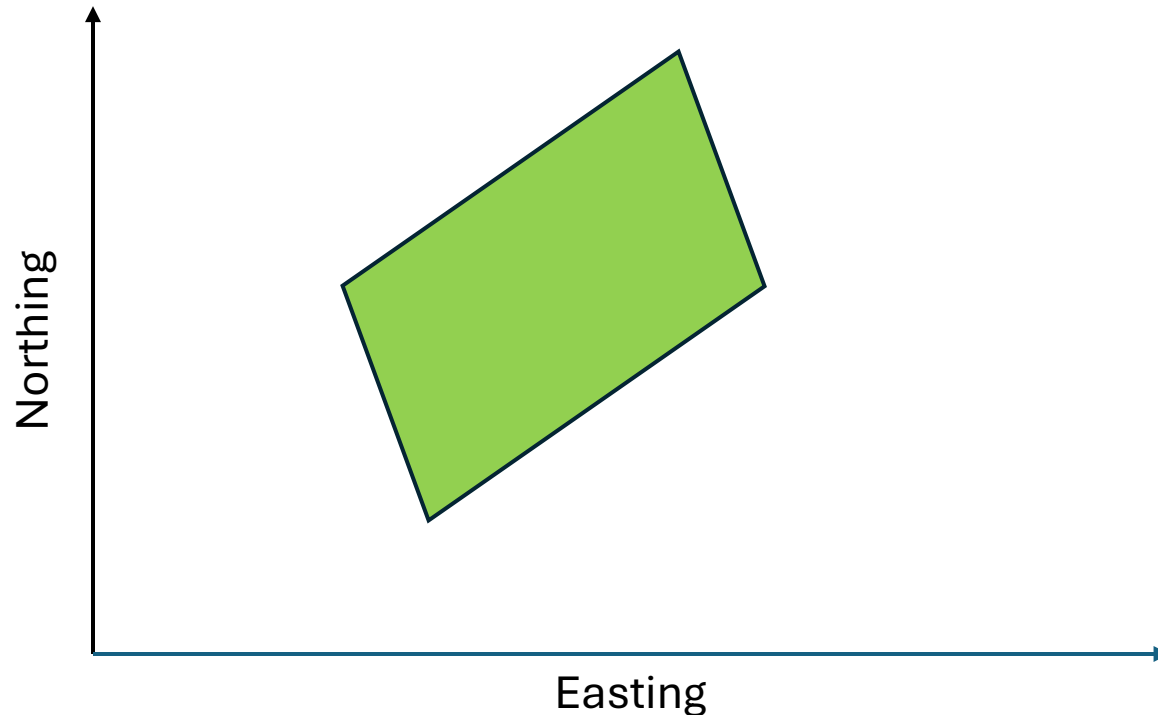


NW	N	NE
W		E
SW	S	SE

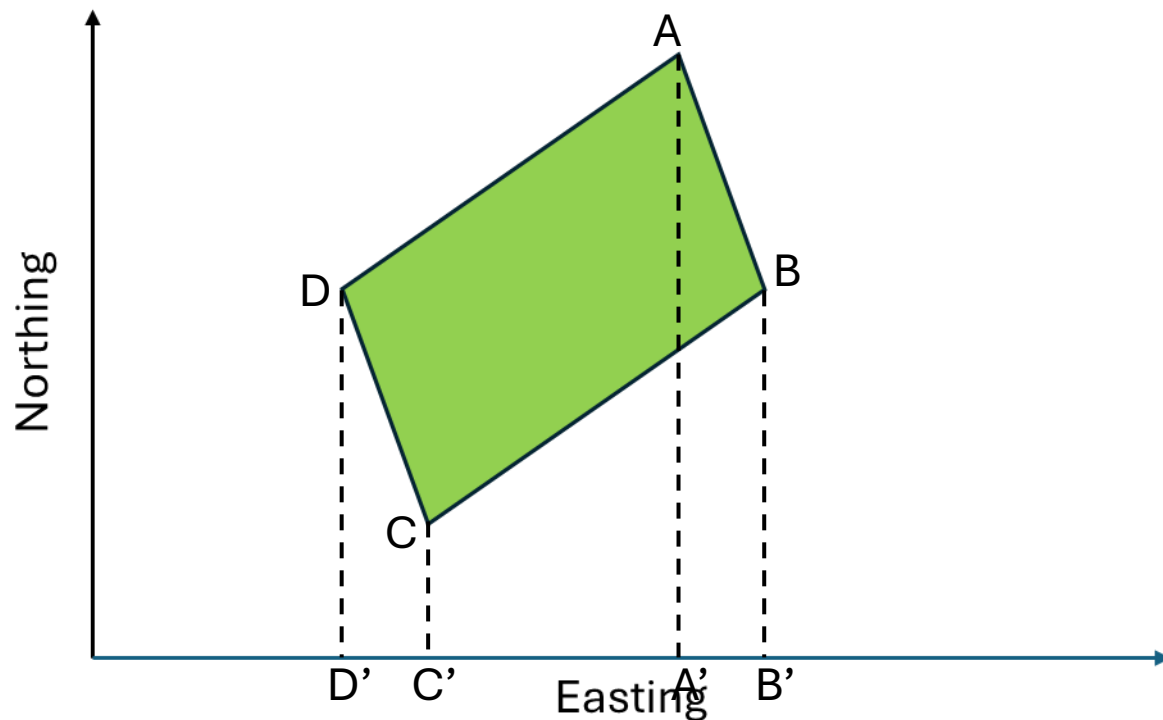
$$Azimuth = \frac{180}{\pi} \tan^{-1} \left(\frac{\Delta H_E}{\Delta H_N} \right)$$

Area

- Physical size of a polygon encompassed by its perimeter
- Defined as the interior extent bounded by an enclosed border



How does computer compute an area?



$$\begin{aligned} \text{Area}(ABCD) = & [\text{Area}(ABB'A') + \text{Area}(AA'D'D)] \\ & - \\ & [\text{Area}(CBB'C') + \text{Area}(CC'D'D)] \end{aligned}$$

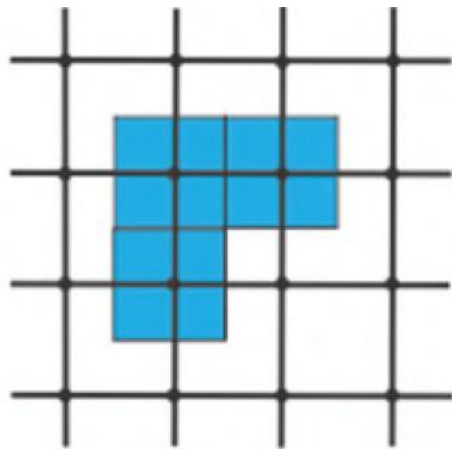
Spatial Tessellation

- Process of partitioning 2D space into a mosaic of very small blocks
- Preparatory step to represent a surface
- Useful in the concept of rasterization

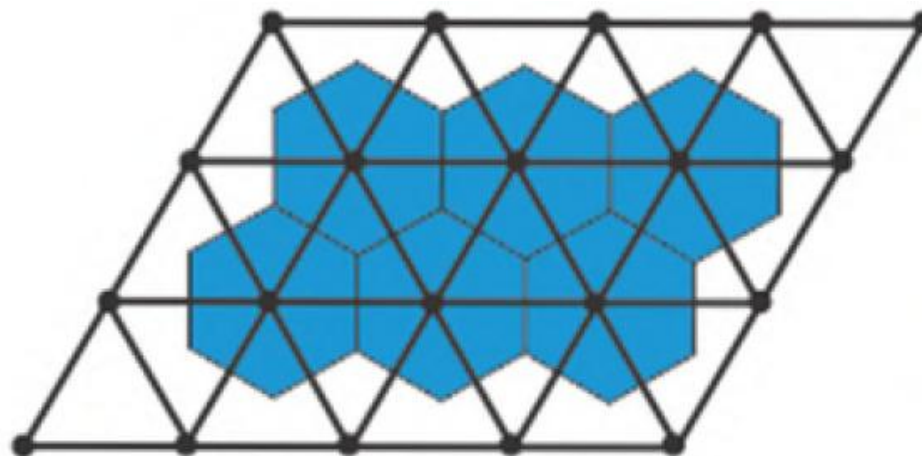
Spatial Tesselation (contd.)

Regular tessellation

- Space is partitioned into regular lattices of squares, triangles, hexagons or compact shapes, all having an equal size
- Depicted attribute value is available at a uniform spatial interval



Regular square lattice

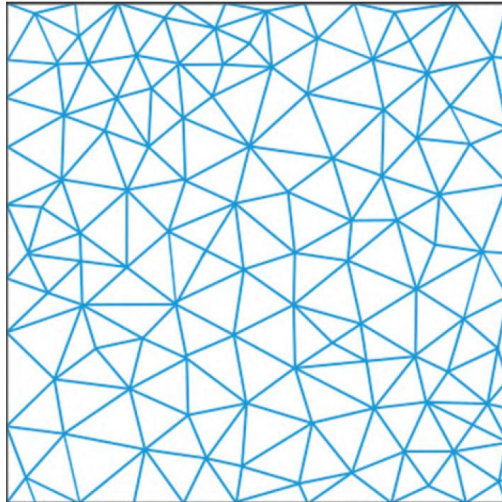


Isosceles triangle lattice

Spatial Tessellation (contd.)

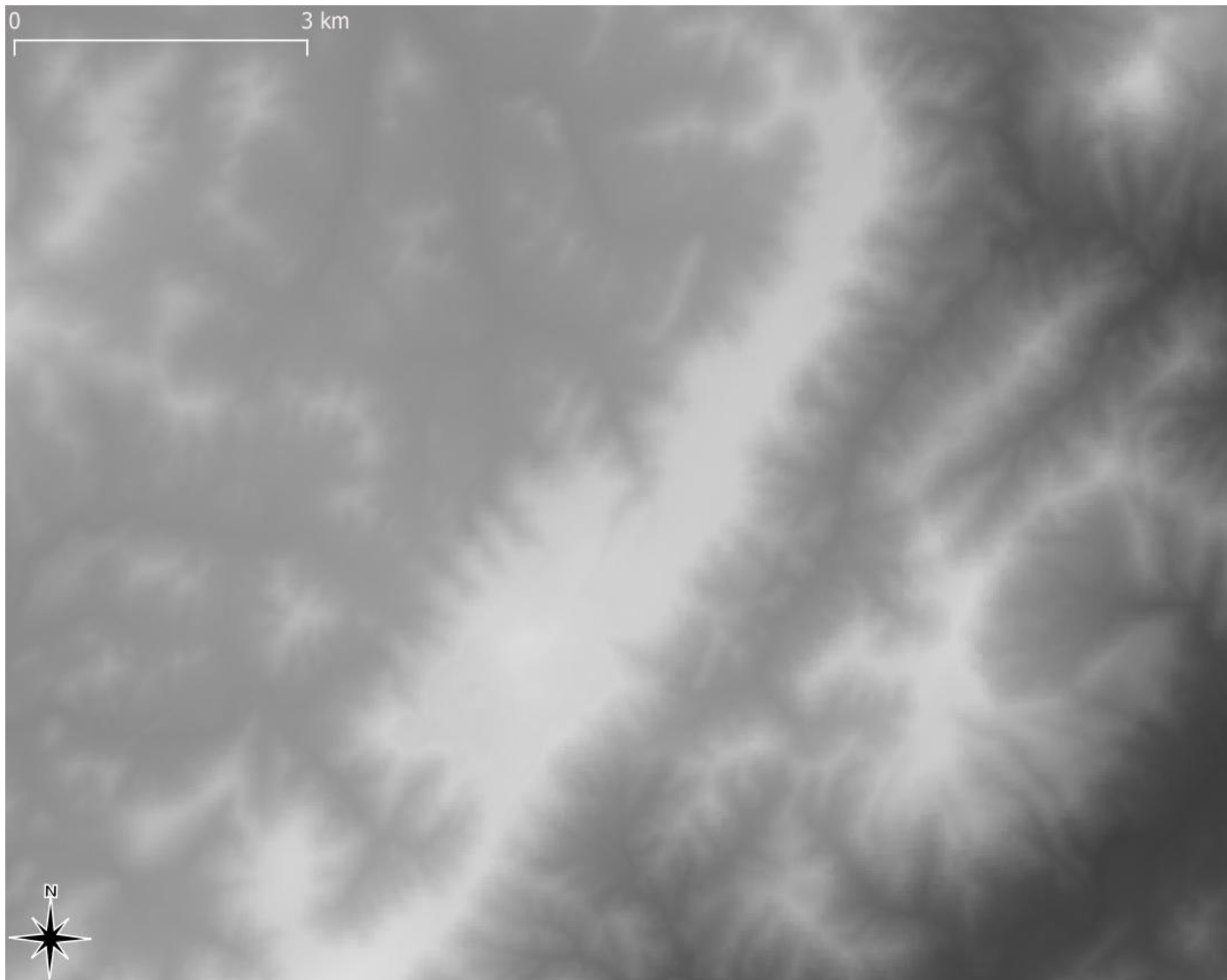
Irregular tessellation

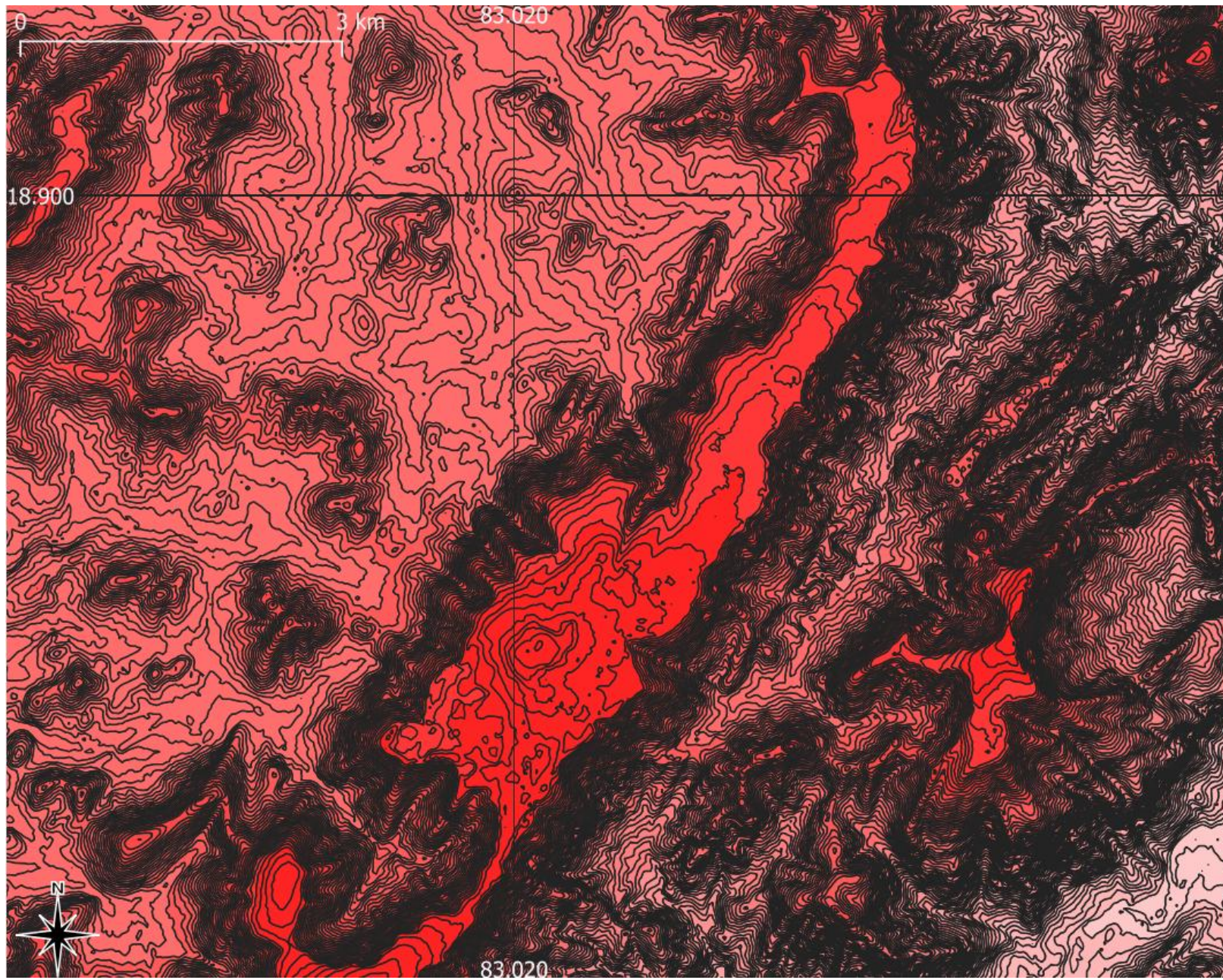
- Space is partitioned into tessellated units of different sizes, but having a similar shape
- E.g., Triangulated Irregular Network (TIN)



Triangulated irregular network: A network of triangles of varying size and orientation. Useful for representing topography

INTERPOLATION





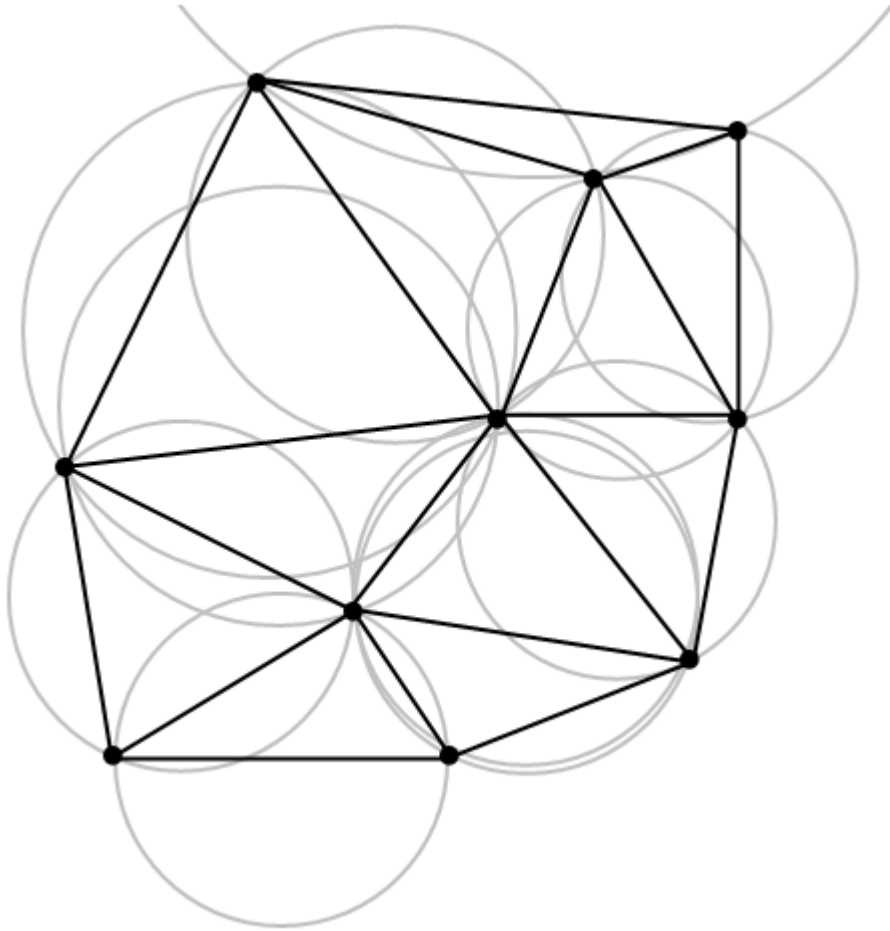
Interpolation:

Estimating values at points intermediate between known data points

- Nearest neighbors
- Inverse distance weighting
- Triangulation
- Natural neighbors
- Kriging

Triangulation

Delaunay triangles

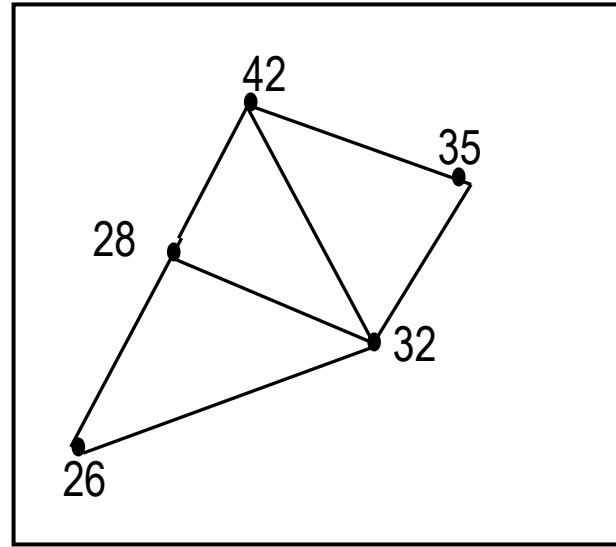
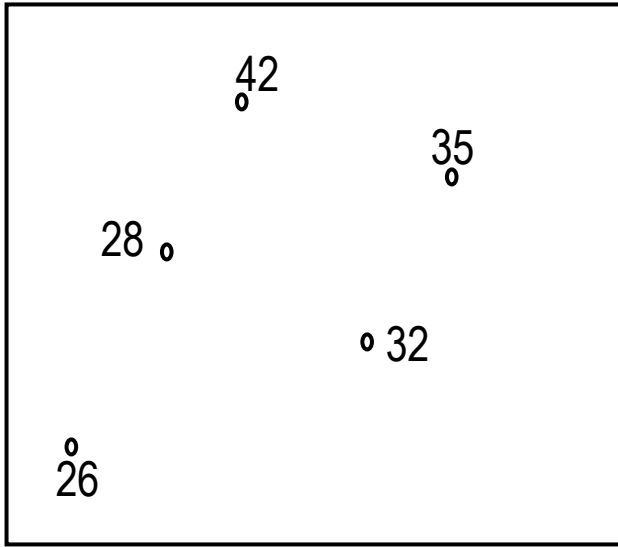


A **Delaunay triangulation** for a set of points is a triangulation of the points in such a way that no point is inside the circumcircle of any triangle.

Delaunay triangulations maximize the minimum angle of all the angles of the triangles in the triangulation.

Triangulation

Draw Delaunay triangles for all points



Value at a point is a function of its location:

$$\Rightarrow z = f(x,y)$$

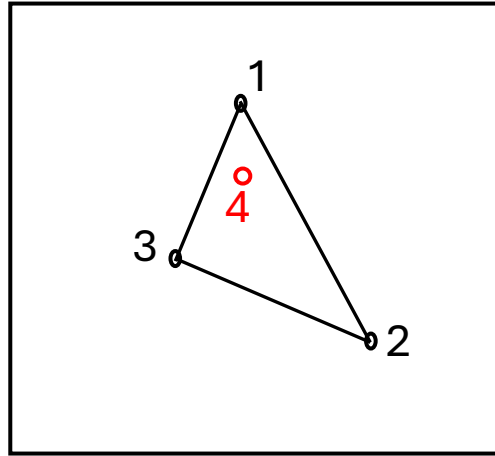
$$z = ax + by + c$$

where z is the value,

x and y are X and Y coordinates, respectively

a , b and c are unknown coefficients

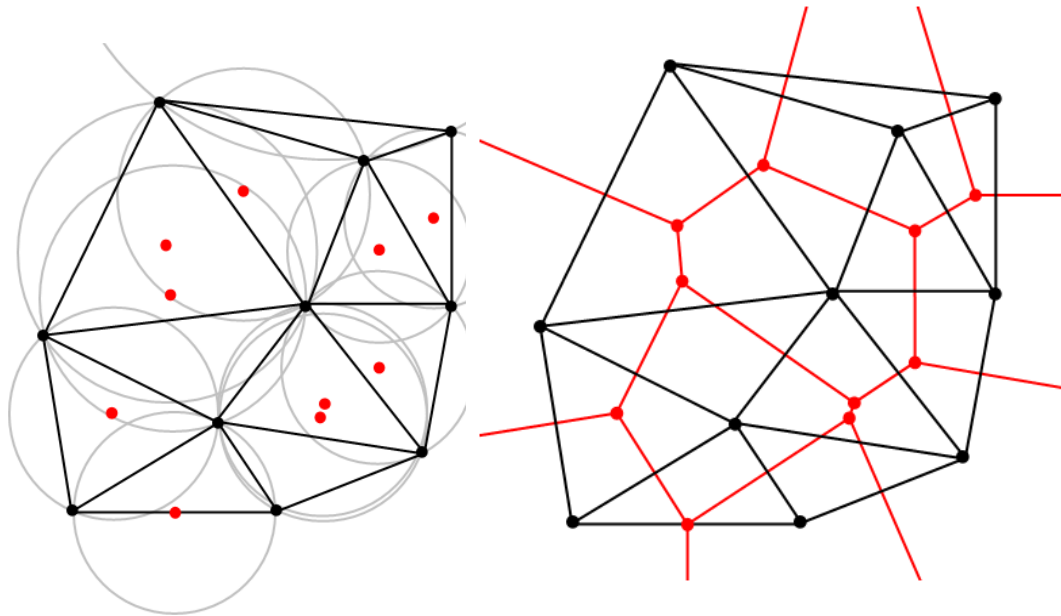
Exercise – Calculate value at point #4



FID	X	Y	Z
1	3	5	42
2	4	2	32
3	2	3	28
4	3	4	?

Natural neighbor

Voronoi polygon

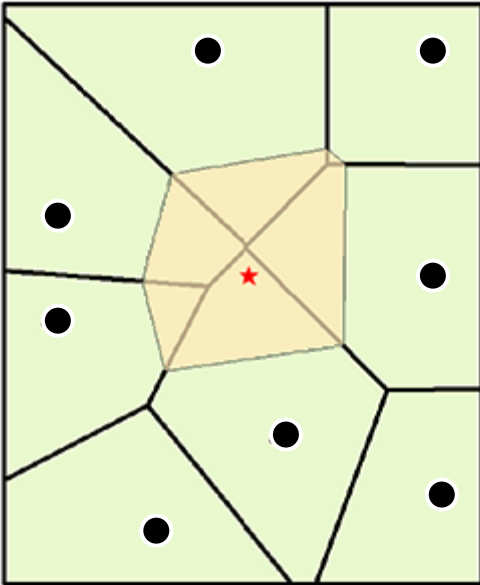


Connecting the centres of the circumcircles produces the Voronoi polygons.

The property of a Voronoi polygon of a point is that all points with that polygon are closest to that point.

Natural neighbour

Natural neighbour interpolation finds the closest subset of points with known values for the query point and applies weights to them based on proportionate areas in order to interpolate.

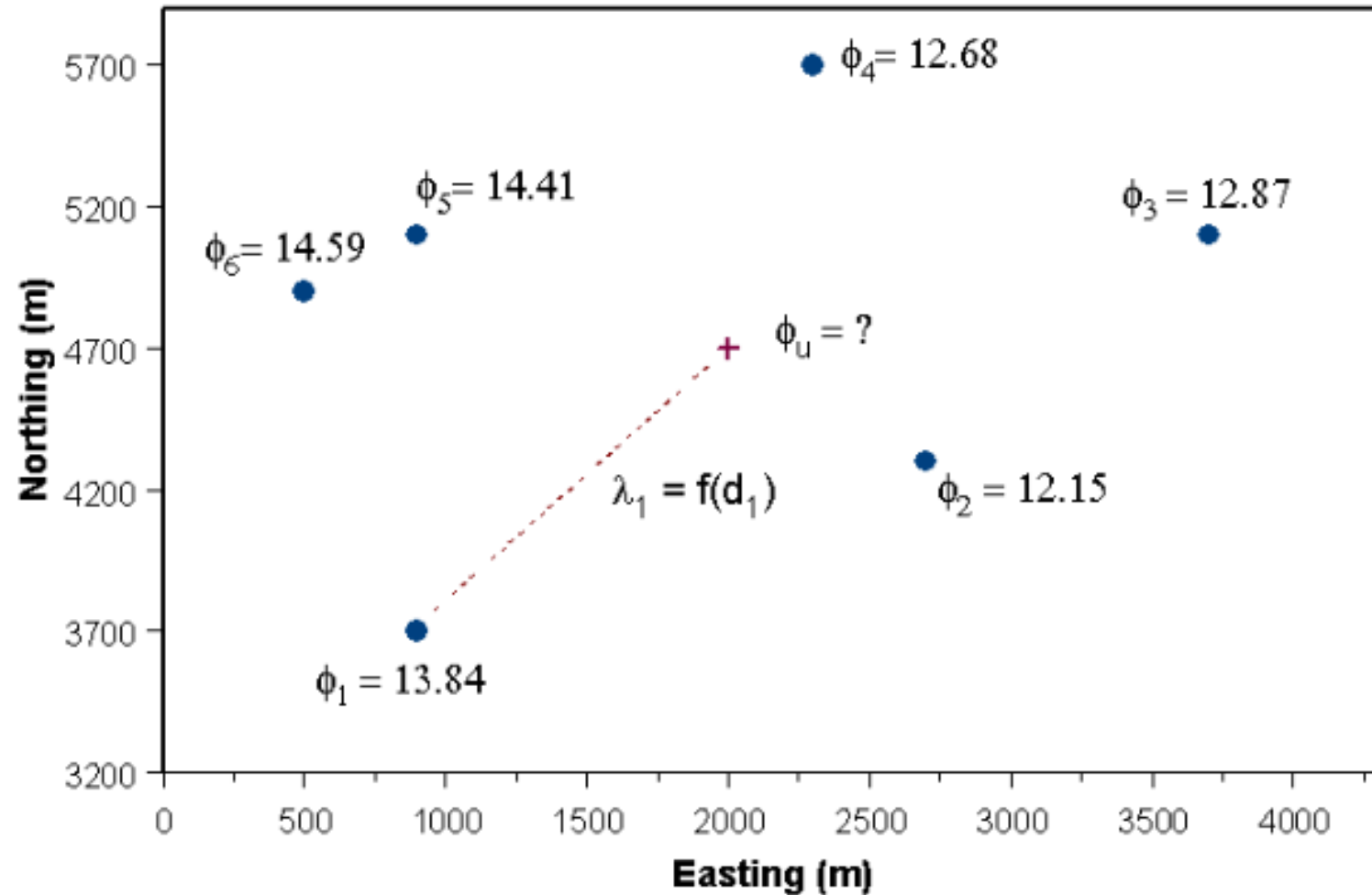


- Draw Voronoi polygons for all points (green colour)
- Draw a Voronoi polygon around the point at which the value is to be determined (orange colour)
- Apply weights to each point value in proportion to the area of intersection between the Voronoi polygon of that point and the the Voronoi polygon of the query point.

Interpolation algorithms

- If the data locations are fairly dense and uniformly distributed throughout the study area, you will get fairly good estimates regardless of interpolation algorithm.
- If the data locations fall in a few clusters with large gaps in between, you will get unreliable estimates regardless of interpolation algorithm.
- Almost all interpolation algorithms will underestimate the highs and overestimate the lows; this is inherent to averaging and if an interpolation algorithm didn't average we wouldn't consider it reasonable

Interpolation



ϕ_u is estimated by a weighted average $\sum \lambda_\alpha \phi_\alpha$, with weights λ_α given by some decreasing function of distance d_α , from u to the data point α

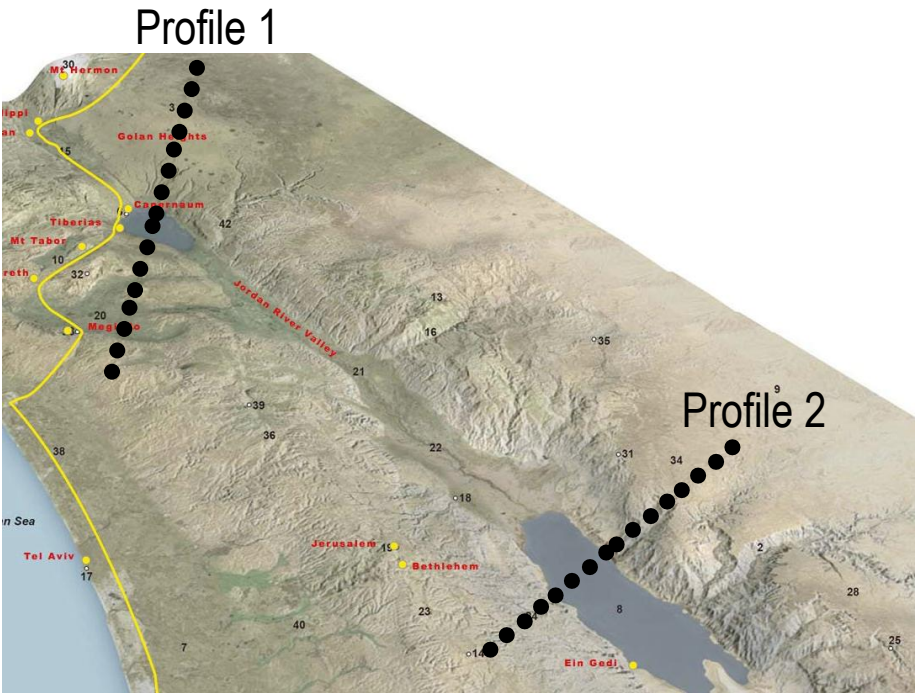
Kriging

- Kriging is a method of spatial interpolation that originated in the field of mining geology as is named after South African mining engineer Danie Krige
- It is one of several methods that use a limited set of sampled data points to estimate the value of a variable over a continuous spatial field

Kriging: Determining topographic profiles



Kriging: Determining topographic profiles

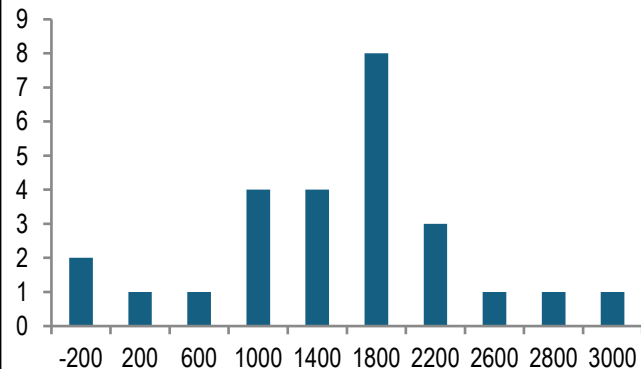


Profile 1		Profile 2	
0	1850	0	2810
1	1700	1	2000
2	1600	2	-200
3	1500	3	2750
4	1550	4	2000
5	1700	5	2500
6	1800	6	-210
7	2000	7	1700
8	1500	8	1700
9	1100	9	1550
10	800	10	1850
11	700	11	1700
12	200	12	1600
13	-210	13	1800
14	-200	14	200
15	400	15	400
16	800	16	700
17	1000	17	800
18	1200	18	800
19	1400	19	1400
20	1300	20	1100
21	1700	21	1500
22	2000	22	1300
23	2500	23	1000
24	2810	24	1200
25	2750	25	1500

Topographic Profiles

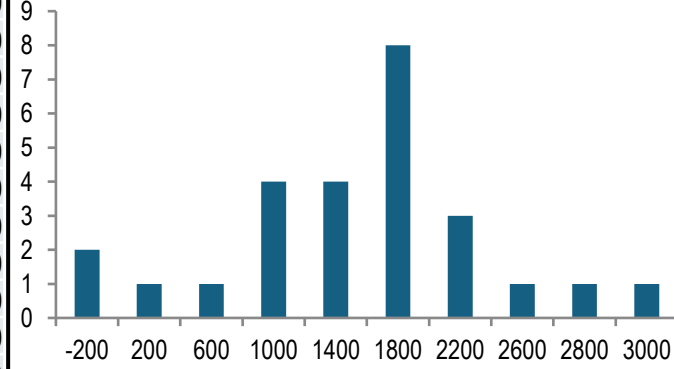
Profile 1	
0	1850
1	1700
2	1600
3	1500
4	1550
5	1700
6	1800
7	2000
8	1500
9	1100
10	800
11	700
12	200
13	-210
14	-200
15	400
16	800
17	1000
18	1200
19	1400
20	1300
21	1700
22	2000
23	2500
24	2810
25	2750

Mean: 1363.462
Stdev: 772.6824
Max: 2810
Min: -210

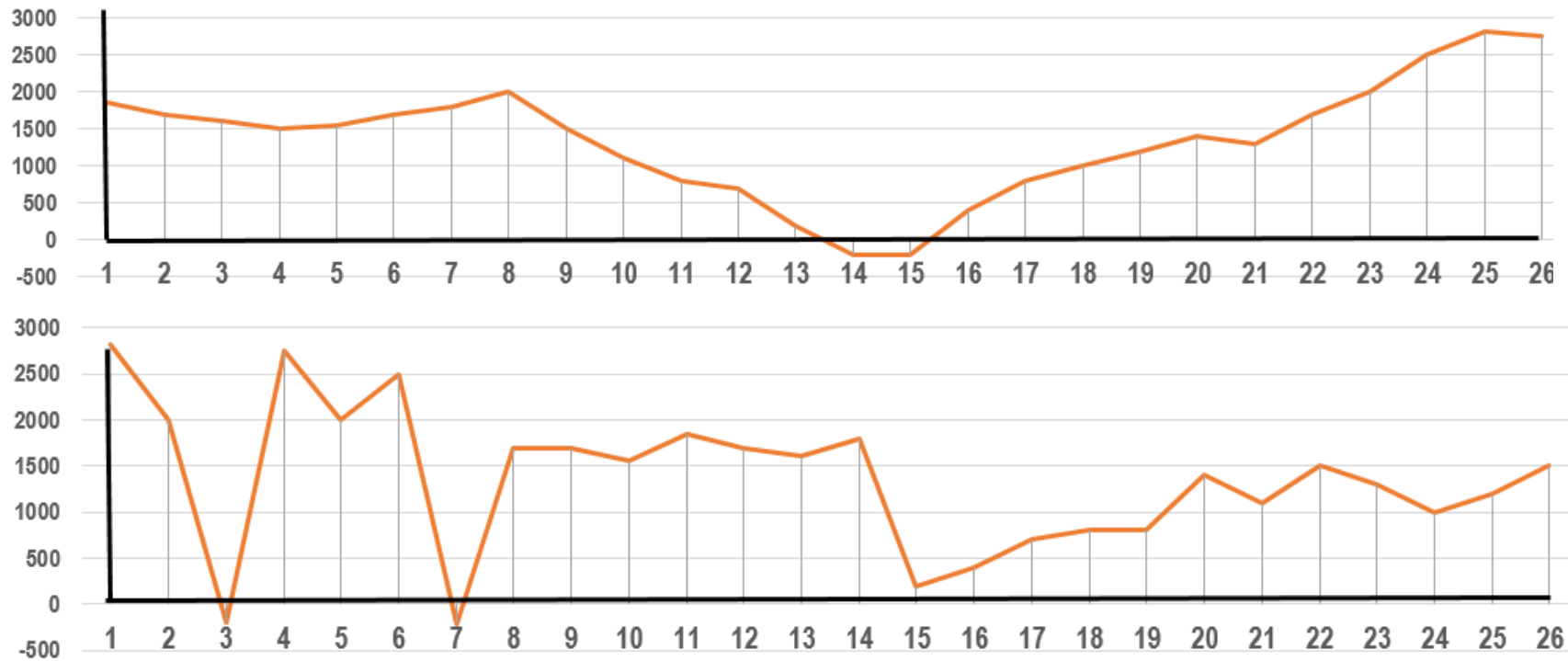


Profile 2	
0	2810
1	2000
2	-200
3	2750
4	2000
5	2500
6	-210
7	1700
8	1700
9	1550
10	1850
11	1700
12	1600
13	1800
14	200
15	400
16	700
17	800
18	800
19	1400
20	1100
21	1500
22	1300
23	1000
24	1200
25	1500

Mean: 1363.462
Stdev: 772.6824
Max: 2810
Min: -210



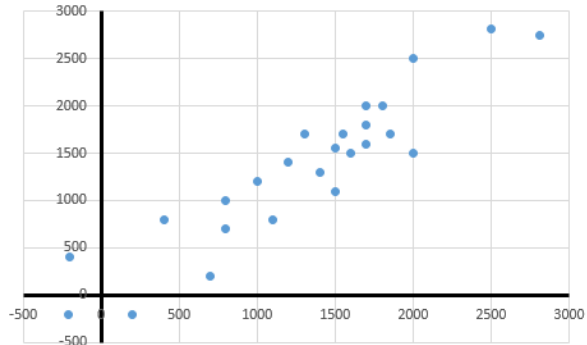
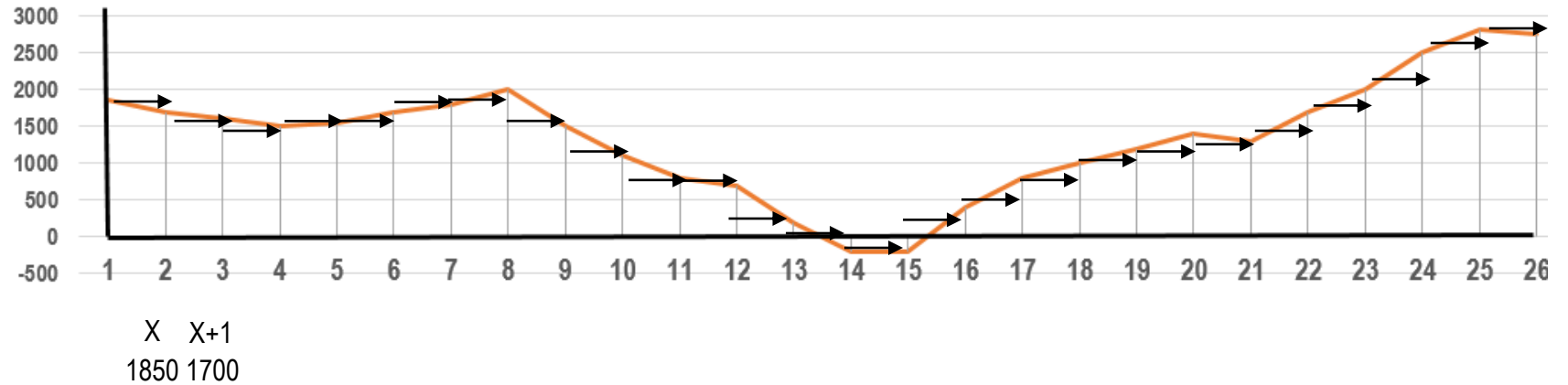
Are the two traverses same in terms of topography?



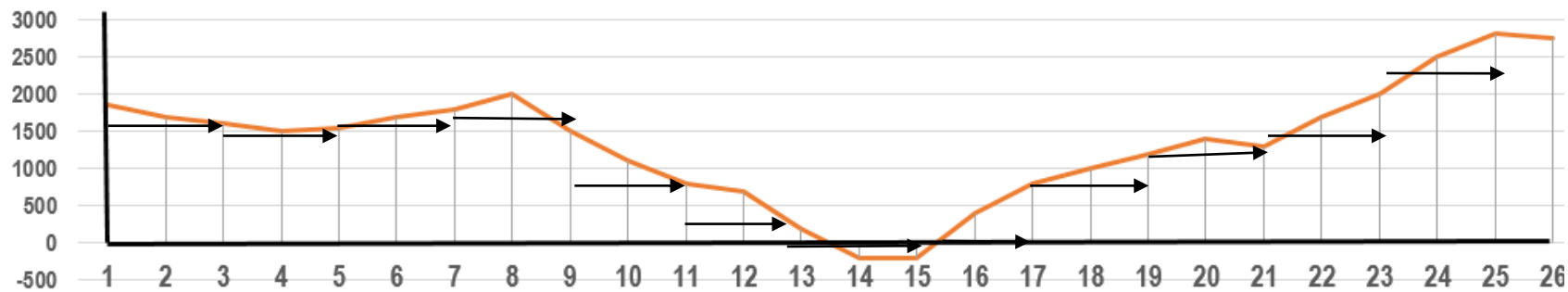
- The profiles are different
- Univariate statistics cannot detect the difference
- Additional statistics is needed

Profile 1

1700 1600
1600 1500
1500 1550
1550 1700
1700 1800
1800 2000
2000 1500
1500 1100
1100 800
800 700
700 200
200 -210
-210 -200
-200 400
400 800
800 1000
1000 1200
1200 1400
1400 1300
1300 1700
1700 2000
2000 2500
2500 2810
2810 2750

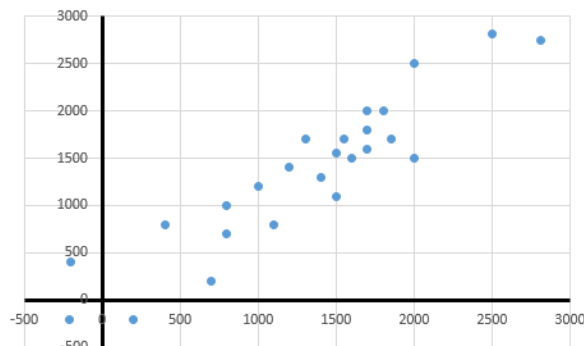


Profile 1



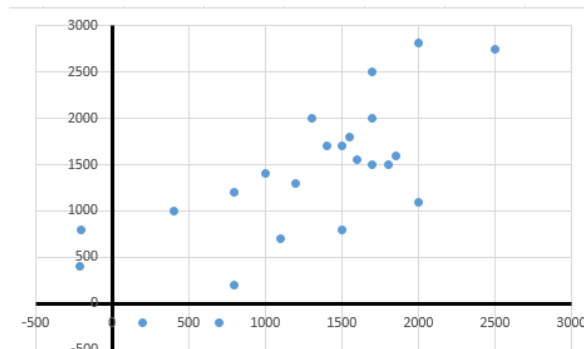
X X+1

1850 1700
1700 1600
1600 1500
1500 1550
1550 1700
1700 1800
1800 2000
2000 1500
1500 1100
1100 800
800 700
700 200
200 -210
-210 -200
-200 400
400 800
800 1000
1000 1200
1200 1400
1400 1300
1300 1700
1700 2000
2000 2500
2500 2810
2810 2750

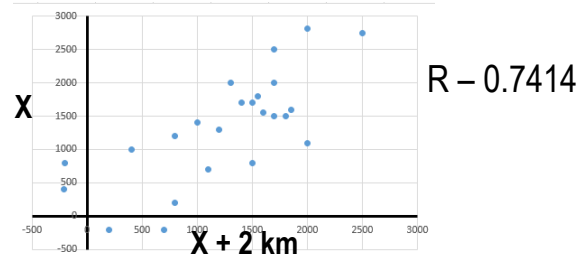
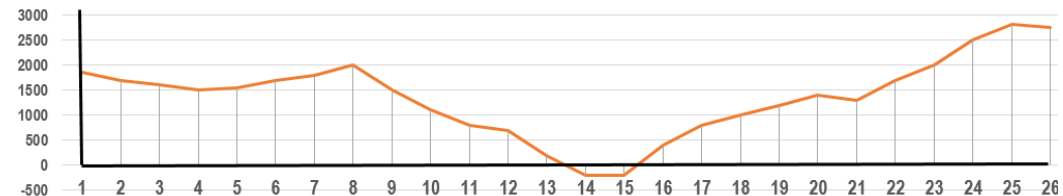
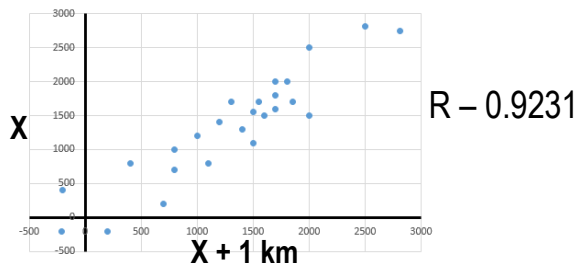


X X+2

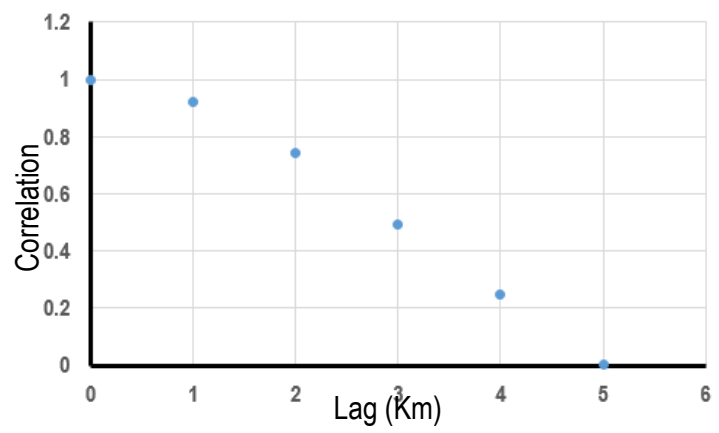
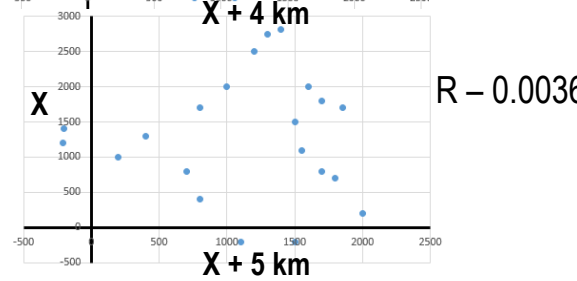
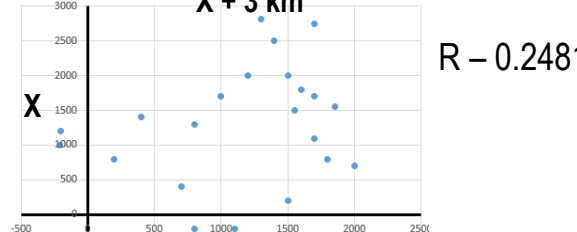
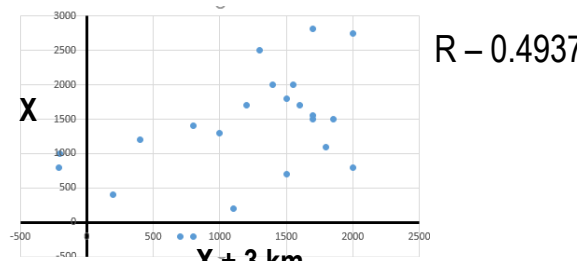
1850 1600
1600 1550
1500 1700
1550 1800
1700 2000
1800 1500
2000 1100
1500 800
1100 700
800 200
700 -210
200 -200
-210 400
-200 800
400 1000
800 1200
1000 1400
1200 1300
1400 1700
1300 2000
1700 2500
2000 2810
2500 2750



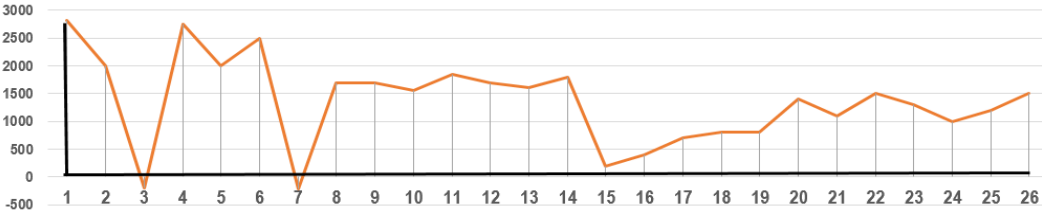
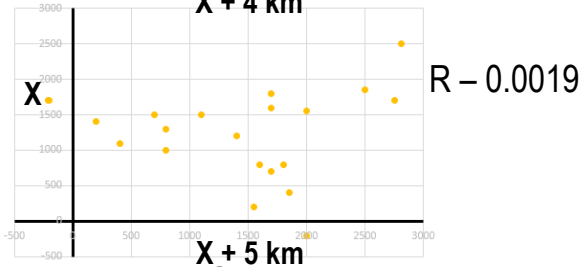
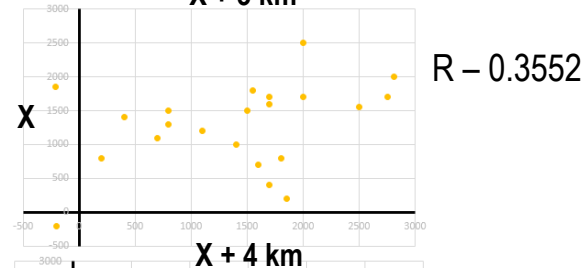
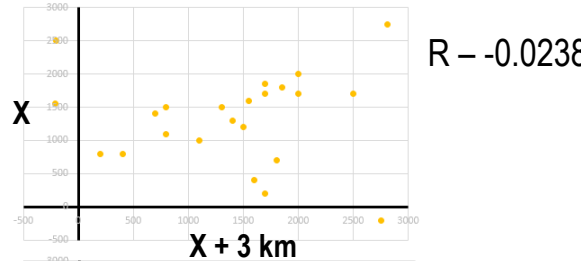
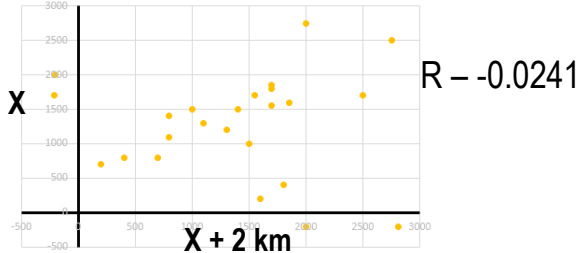
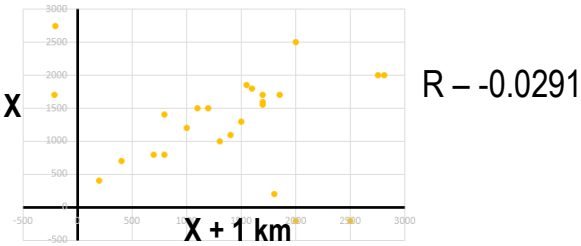
Profile 1



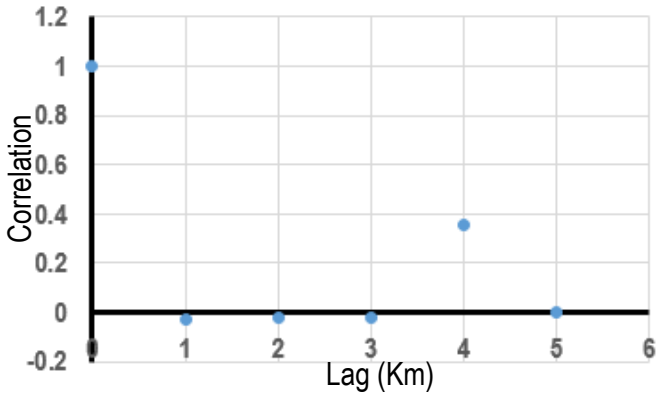
Lag (Km)	R
0	1
1	0.923137
2	0.741453
3	0.493793
4	0.248143
5	0.003663



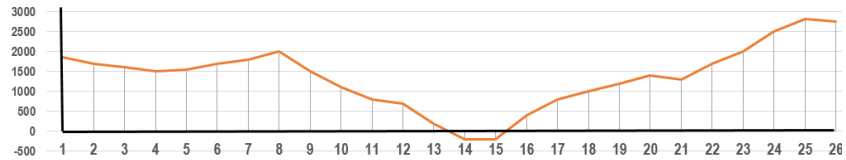
Profile 2



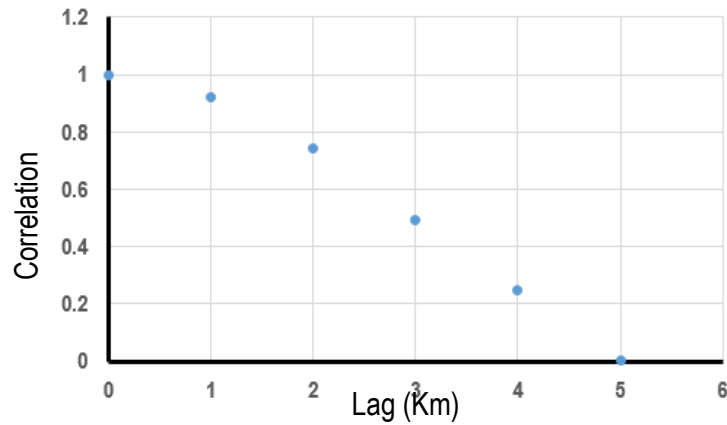
Lag (Km)	R
0	1
1	-0.0291
2	-0.0241
3	-0.0238
4	0.03552
5	0.0019



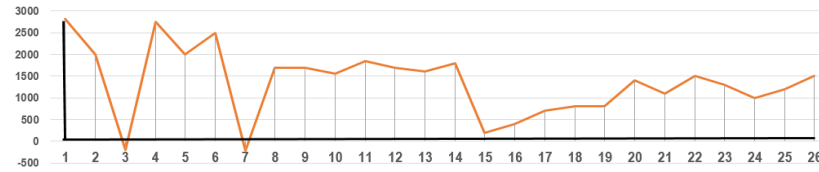
Profile 1



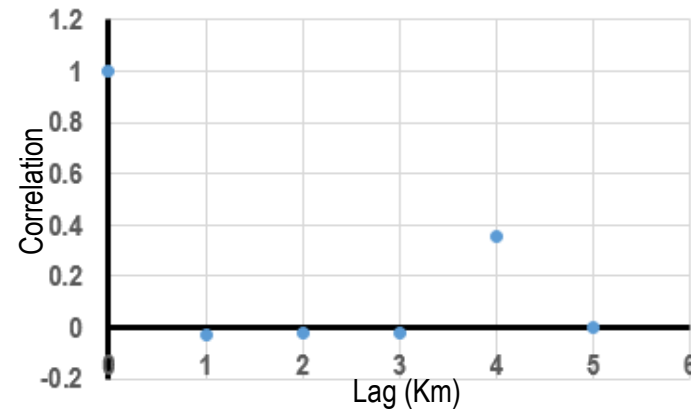
Lag (Km)	R
0	1
1	0.923137
2	0.741453
3	0.493793
4	0.248143
5	0.003663



Profile 2



Lag (Km)	R
0	1
1	-0.0291
2	-0.0241
3	-0.0238
4	0.03552
5	0.0019



The lag is the distance between sample pairs that a statistics is calculated for.

The Correlogram

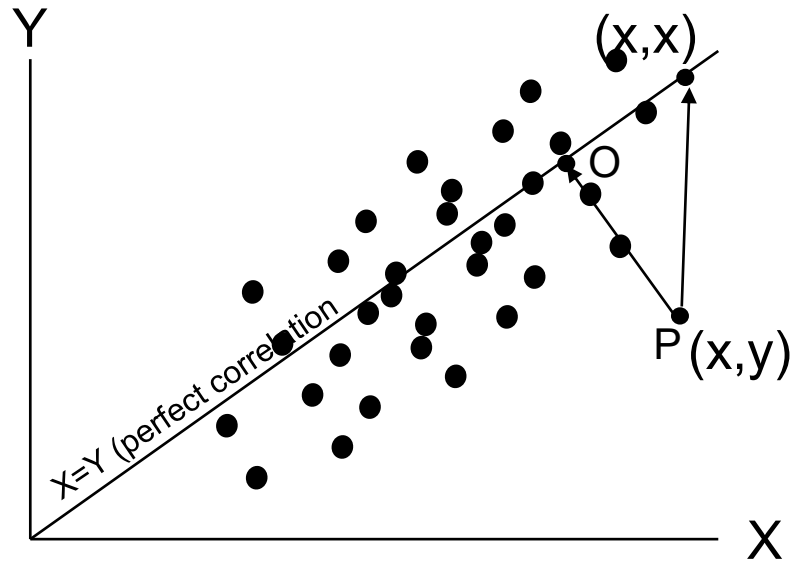
- Correlogram estimates the correlation between two points in space
- The correlation is a function of distance
- The correlation may also be a function of direction
- The separation vector defines both direction and distance between sample points

Other estimators of spatial continuity or variability

- The Correlogram: $r = \frac{\sigma_{xy}}{\sigma_x \sigma_y}$
- The covariance function : $C = \frac{1}{n} \sum_1^n (x_i - \bar{x})(y_i - \bar{y})$
- The Semivariogram

Semi-variogram of Point P

Can be seen as Moment of Inertia of the Point P
(Mass x Distance squared)



$$\frac{OP}{x-y} = \frac{1}{\sqrt{2}}$$

$$OP = \frac{x-y}{\sqrt{2}}$$

$$\text{Moment of inertia} = \frac{(x-y)^2}{2}$$

$$\text{Semivariogram} = \text{Average Moment of Inertia} = \frac{1}{n} \sum_{i=1}^n \frac{(x_i - x_{i+h})^2}{2}$$

$$\text{Semivariogram} = \frac{1}{2n} \sum_{i=1}^n (x_i - x_{i+h})^2, h = \text{lag}$$

Experimental Semivariance

- ... here we call it the experimental semivariance for lag h :

$$\hat{\gamma}(h) = \frac{1}{2N(h)} \sum_{\alpha=1}^{N(h)} [x_o - x_{(o+h)}]^2$$

- $N(h)$ is the number of pairs separated by lag h (+/- lag tolerance)
- h - lag width or the distance between pairs of points
- x - is the value of a point at “ o ” or “ $o + h$ ”

Semivariance, Covariance and Correlation Spread vs Similarity

- **Expectation**, $E(X)$, is the outcomes of a *Random Variable* weighted by their probability.
- **Variance** ($Var = E(X^2) - E(X)^2$) is the difference between *Expectation* of a squared *Random Variable* and the *Expectation* of that *Random Variable* squared
- **Covariance** ($\gamma = E(XY) - E(X)E(Y)$) is the same as *Variance*, only two *Random Variables* are compared, rather than a single *Random Variable* against itself.
- **Correlation**, $\left(R = \frac{C(X,Y)}{\sqrt{Var(X)Var(Y)}} \right)$, is just the *Covariance* normalized.

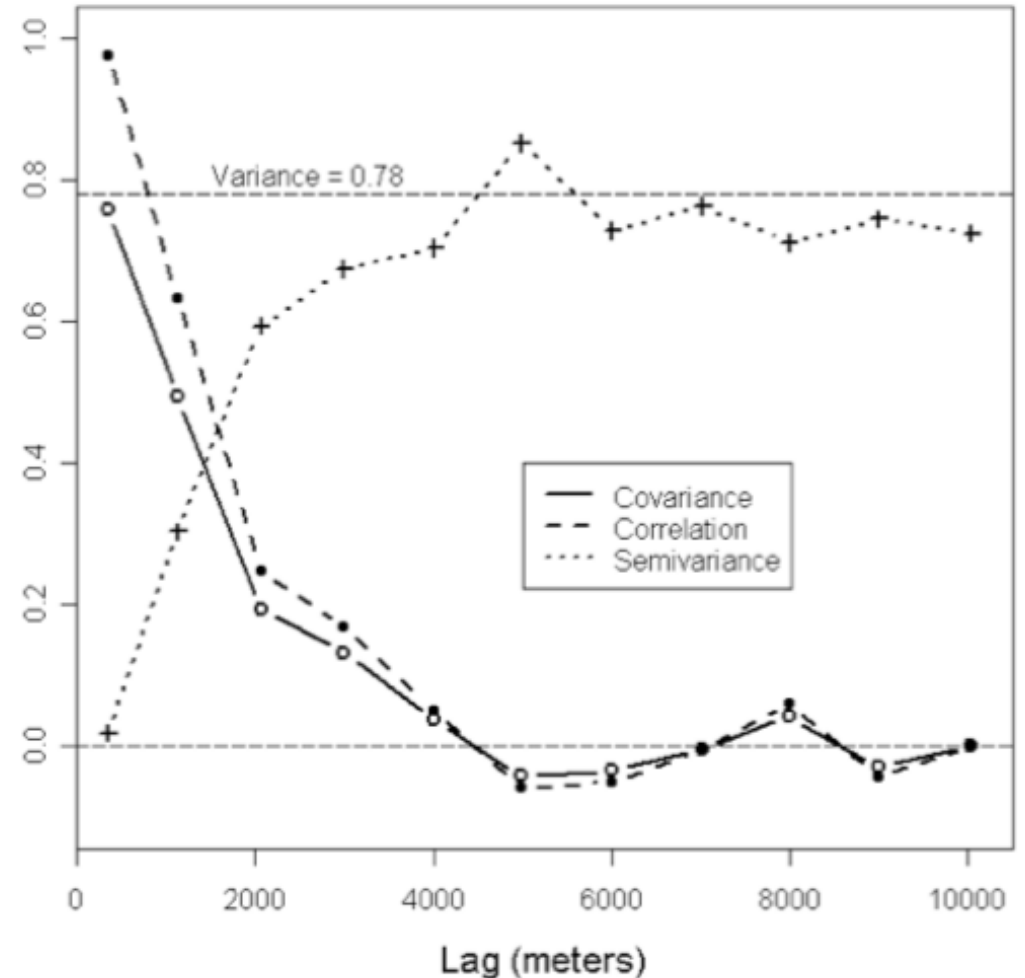
Semivariance, Covariance and Correlation

Spread vs Similarity

$$\hat{\gamma}(\vec{h}) = \frac{1}{2N(\vec{h})} \sum_{\alpha=1}^{N(h)} [x_o - x_{(o+\vec{h})}]^2$$

$$\hat{C}(\vec{h}) = \frac{1}{N(\vec{h})} \sum_{\alpha=1}^{N(h)} (x_o - \bar{x})(x_{(o+\vec{h})} - \bar{x})$$

$$\hat{\rho}(\vec{h}) = \frac{C(h)}{\sqrt{\sigma_o * \sigma_{o+h}}}$$

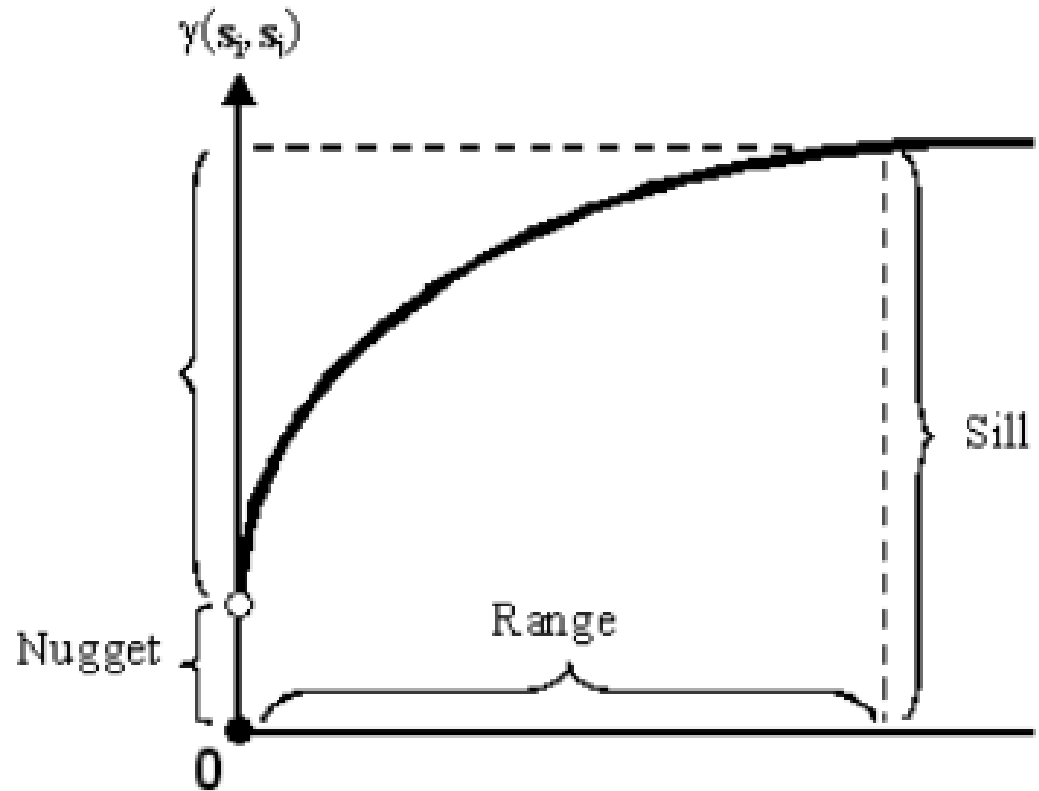


Semivariogram Model

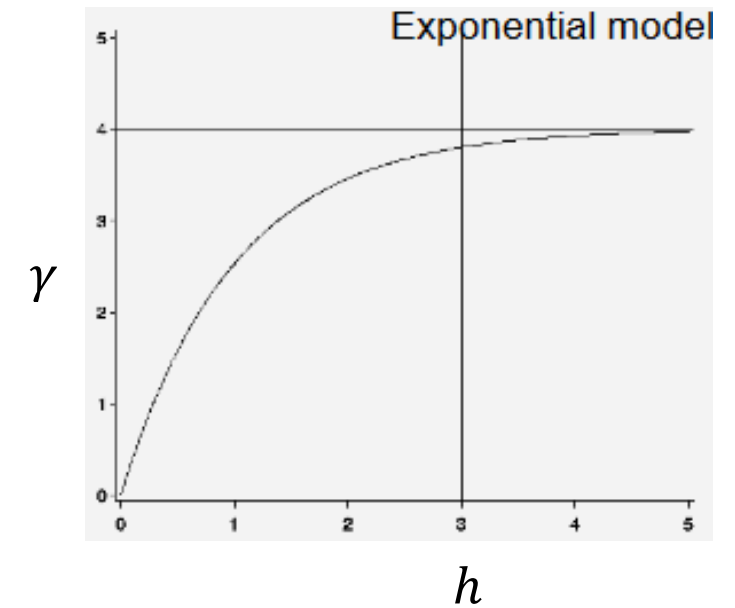
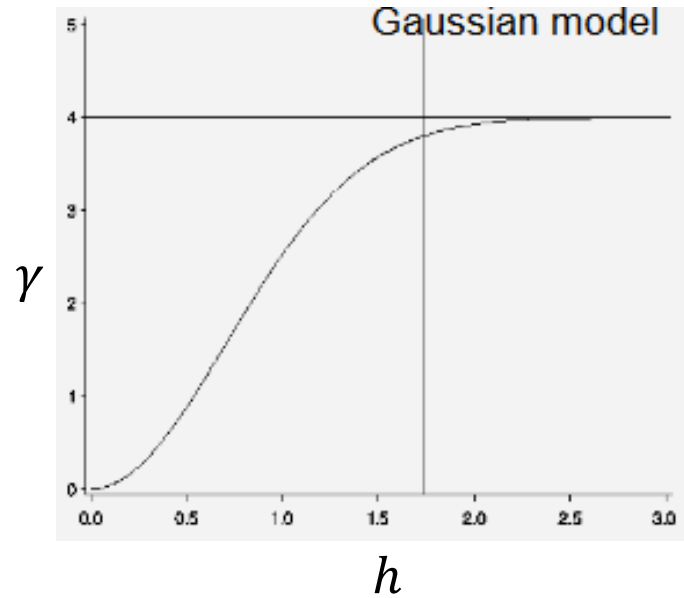
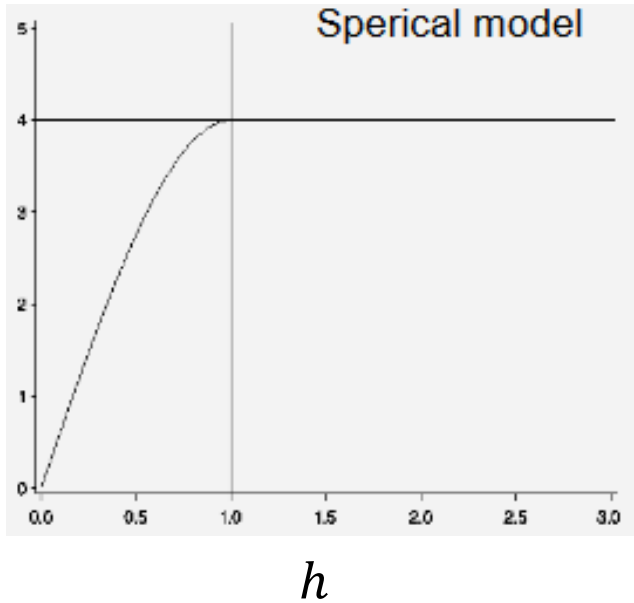
- The empirical semivariogram allows us to derive a semivariogram model to represent semivariance as a function of separation distance
- Thus we can infer the characteristics of the underlying process using the model
- Compute the semivariance between points
- Interpolate between sample points using an optimal interpolator (“kriging”)

Constraints for a Semivariogram Model

- Monotonically increasing
- Asymptotic max (sill)
- Non-negative intercept (nugget)
- Anisotropy



Semivariogram Models



$$\gamma(h) = C_o \left[\frac{3}{2} \frac{h}{\alpha_o} - \frac{1}{2} \left(\frac{h}{\alpha_o} \right)^3 \right]$$

$$\gamma(h) = C_o \left[1 - \exp \left(-\frac{h^2}{\alpha_o^2} \right) \right]$$

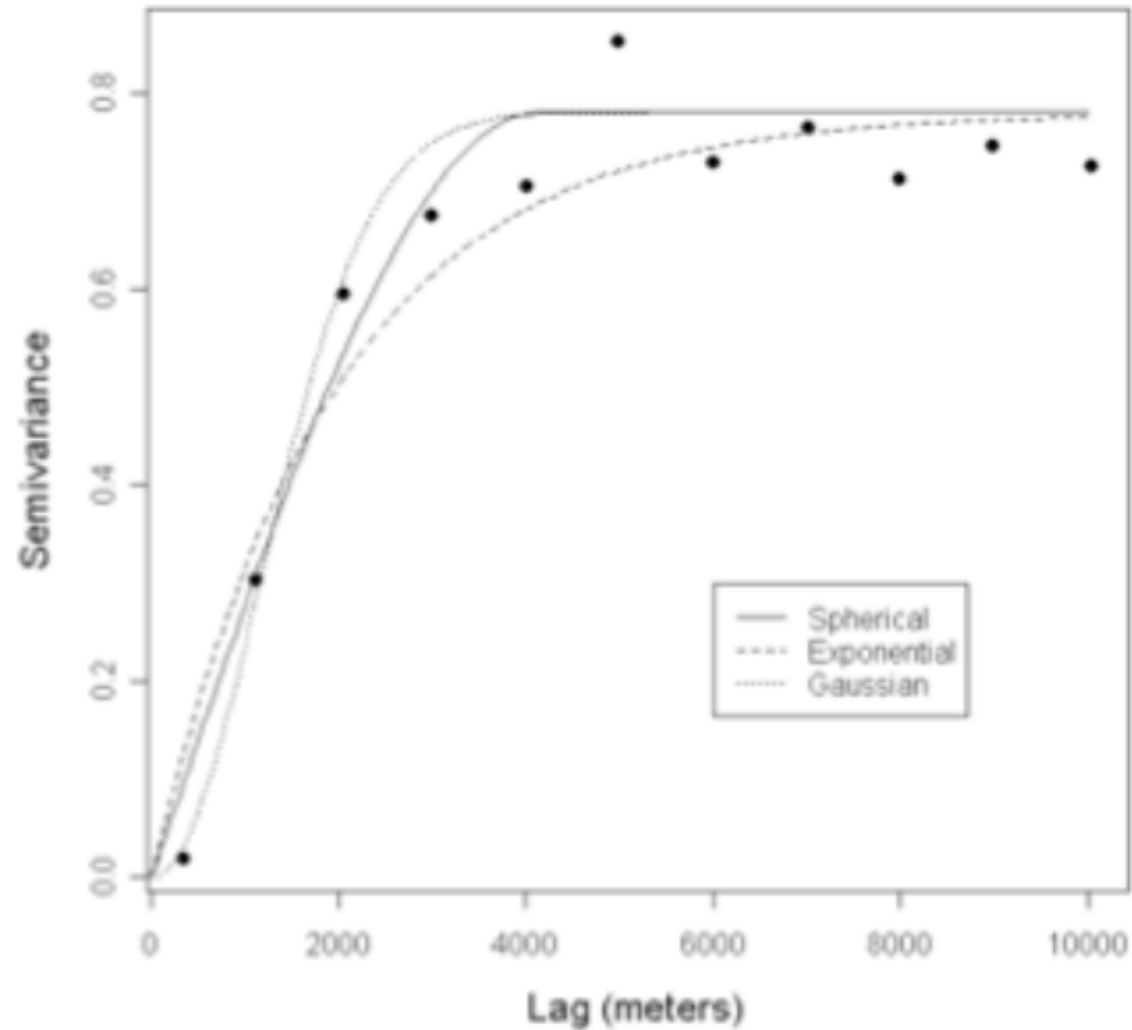
$$\gamma(h) = C_o \left[1 - \exp \left(-\frac{h}{\alpha_o} \right) \right]$$

C_o – Nugget

α_o – Range

What can you think of would happen if there is a trend in the data?

Where is the Best Fit?



Kriging

The value at the queried point is given by:

$$\hat{Y} = \sum w_i Y_i$$

Where Y_i are the values at sample points
 w_i are the weights of sample points

$$C \bullet w = D$$

$$\begin{bmatrix} C_{11} & \dots & C_{1n} & 1 \\ \dots & \dots & \dots & \dots \\ C_{n1} & \dots & C_{nn} & 1 \\ 1 & \dots & 1 & 0 \end{bmatrix} \bullet \begin{bmatrix} w_1 \\ \dots \\ w_n \\ \lambda \end{bmatrix} = \begin{bmatrix} C_{10} \\ \dots \\ C_{n0} \\ 1 \end{bmatrix}$$

$$C^{-1} \bullet C \bullet w = D \bullet C^{-1}$$

C_{11} to C_{nn} – Covariance between the pair of sample points

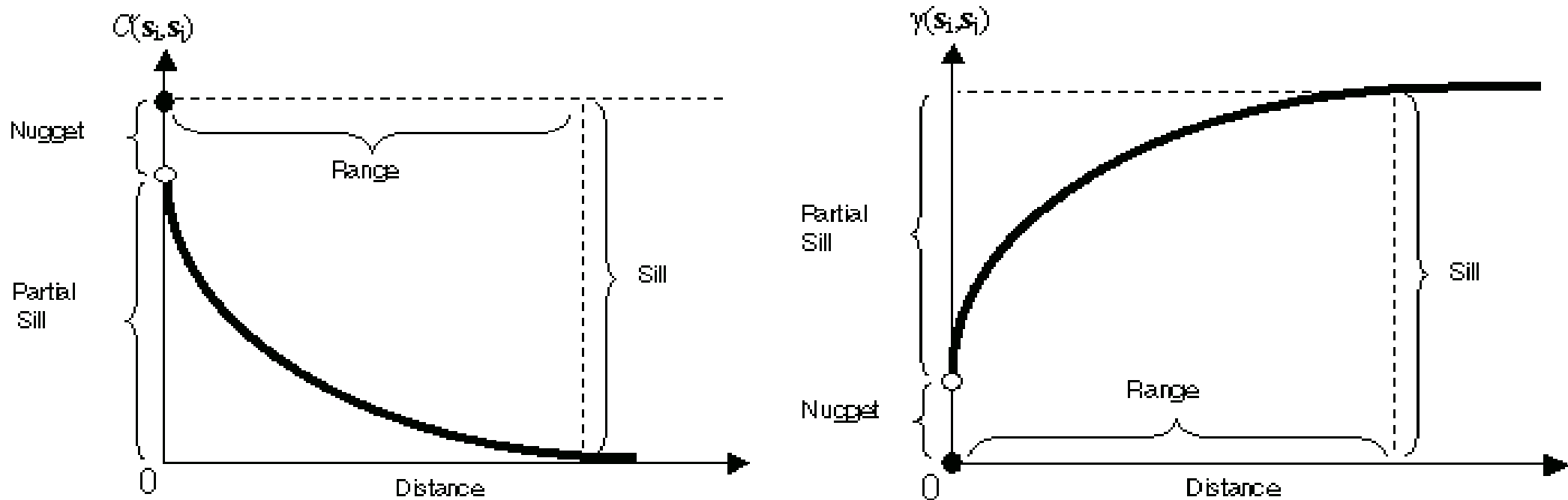
C_{10} to C_{n0} – Covariance between sample points and the point where the value is required to be estimated

Or

$$w = C^{-1} \bullet D$$

Kriging: Equivalence of covariance and semivariogram

Semivariogram can be used to calculate moment of inertia at different distances, hence different C_{ij} in the equations below:



$$\gamma(x_i, x_j) = \text{Sill} - C(x_i, x_j)$$

Kriging: covariance and semivariogram

Semivariogram can be used to calculate moment of inertia at different distances, hence different C_{ij} in the equations below:

$$C \bullet W = D$$
$$\begin{bmatrix} C_{11} & \dots & C_{1n} & 1 \\ \dots & \dots & \dots & \dots \\ C_{n1} & \dots & C_{nn} & 1 \\ 1 & \dots & 1 & 0 \end{bmatrix} \bullet \begin{bmatrix} w_1 \\ \dots \\ w_n \\ \lambda \end{bmatrix} = \begin{bmatrix} C_{10} \\ \dots \\ C_{n0} \\ 1 \end{bmatrix}$$

The following equation is then used to estimate the value at the query point

$$\hat{z} = \sum_{i=1}^n w_i z_i$$